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(54) **THERMOSTABLE VIRAL REVERSE
TRANSCRIPTASE**

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(2013.01); **C12Y 207/07049** (2013.01)

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CPC **C12N 9/1276**
See application file for complete search history.

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(57) **ABSTRACT**

The present invention provides novel engineered reverse
transcriptase enzymes that afford beneficial improvements
in thermal stability, processivity, cDNA yields and elimina-
tion of secondary enzymatic activity. The present invention
also provides methods for amplifying template nucleic acids
using such reverse transcriptase enzymes. This invention
addresses deficiencies in the current state of the art reverse
transcriptase enzymes in RNA detection and analysis includ-
ing deficiencies in detection sensitivity, specificity, side
enzyme activities, enzyme stability and synthesis capacity,
especially when using template nucleic acids ranging in
length, secondary structure and nucleotide content.

24 Claims, 7 Drawing Sheets

Specification includes a Sequence Listing.

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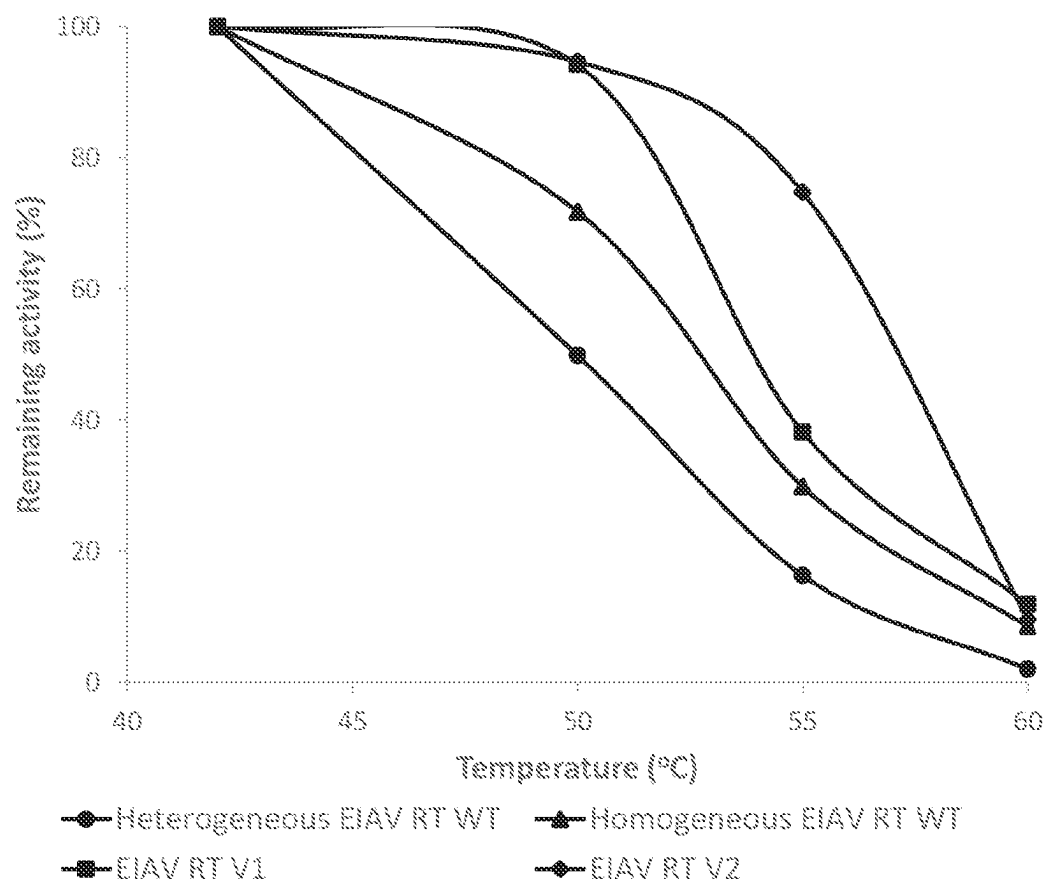


Fig. 1

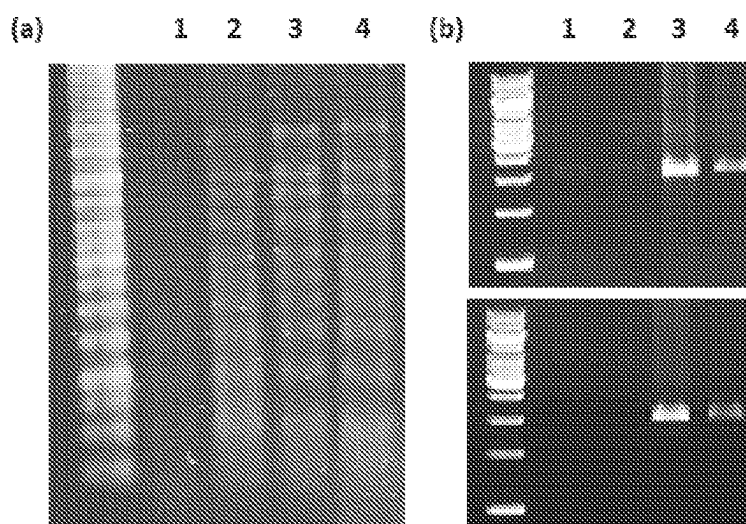


Fig. 2

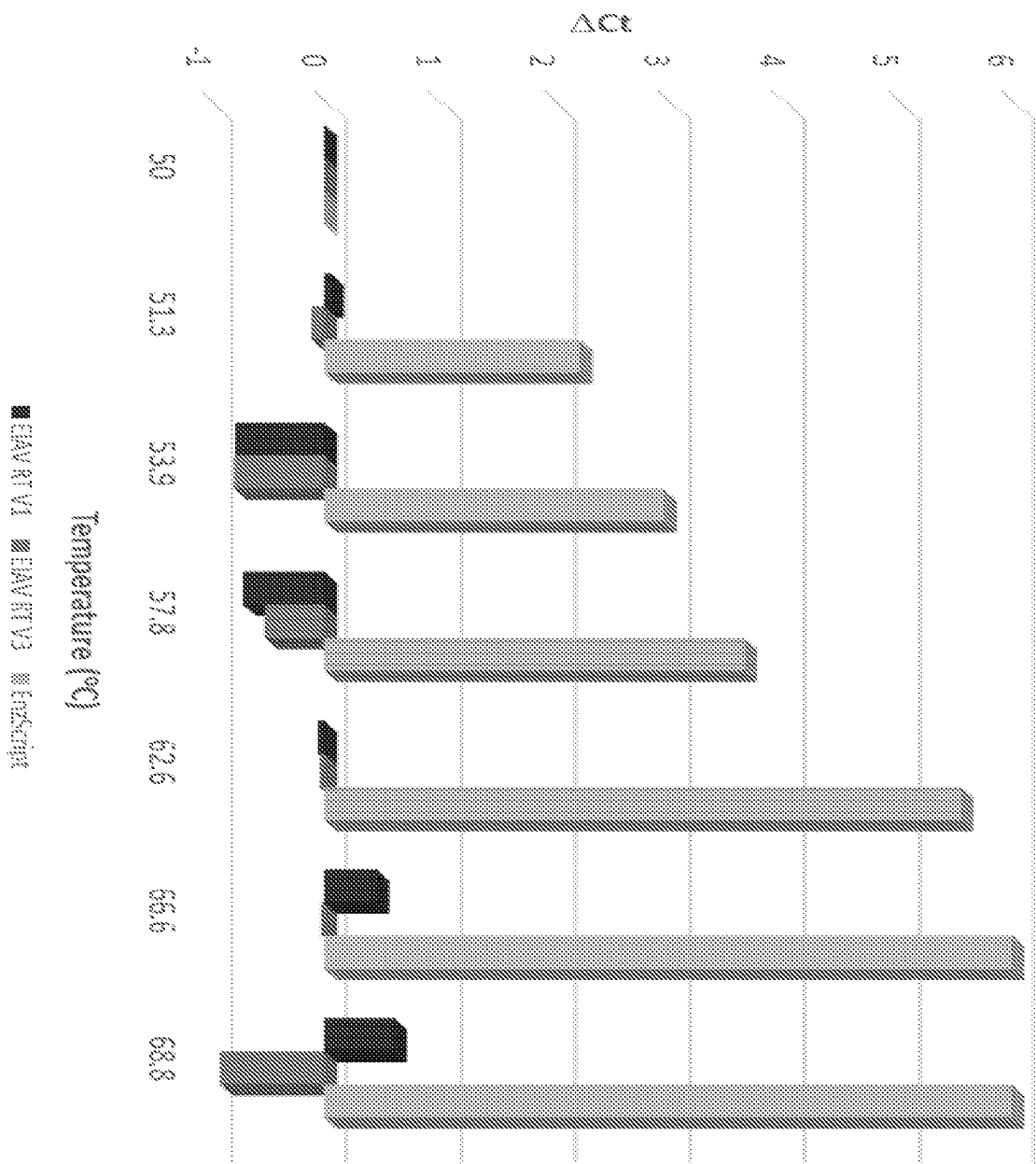


Fig. 3

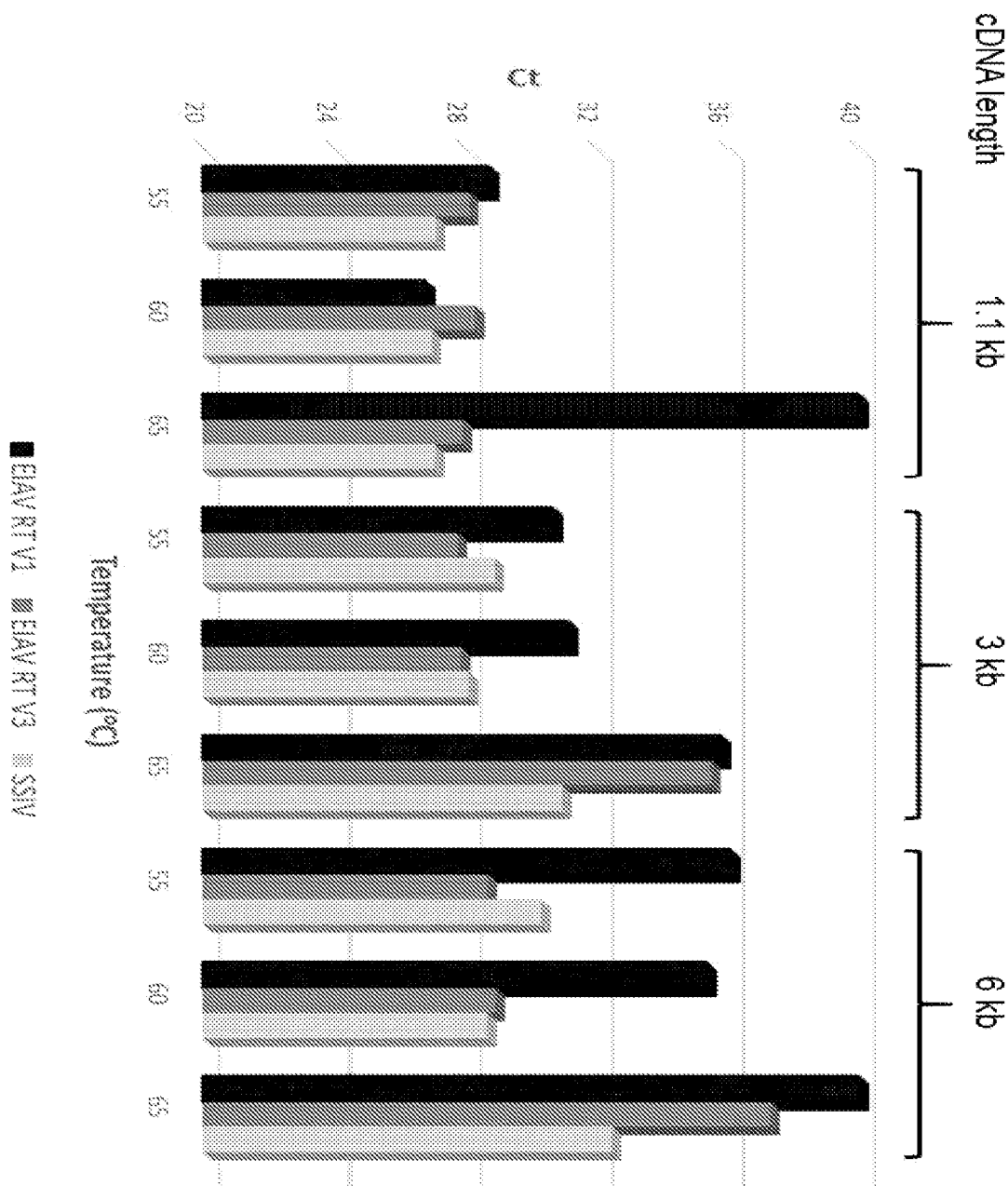


Fig. 4

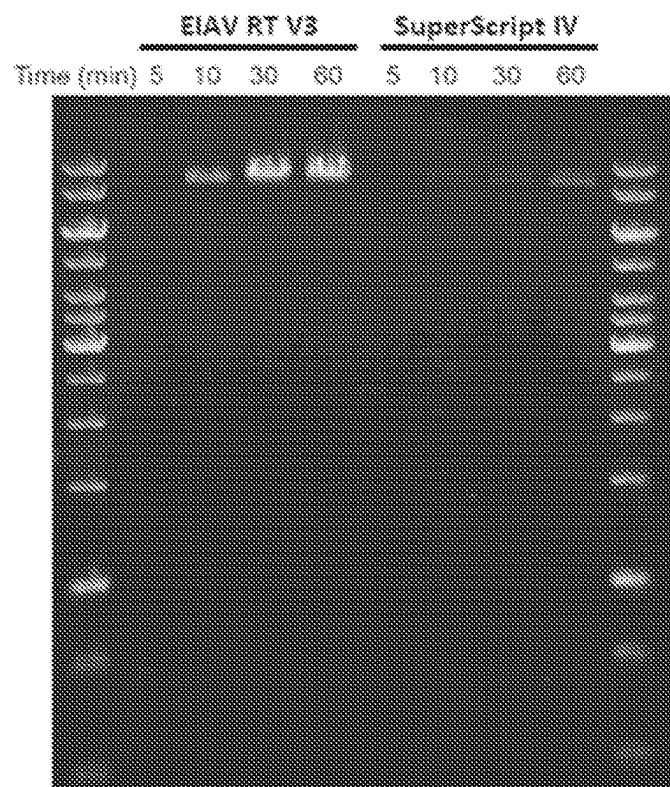


Fig. 5

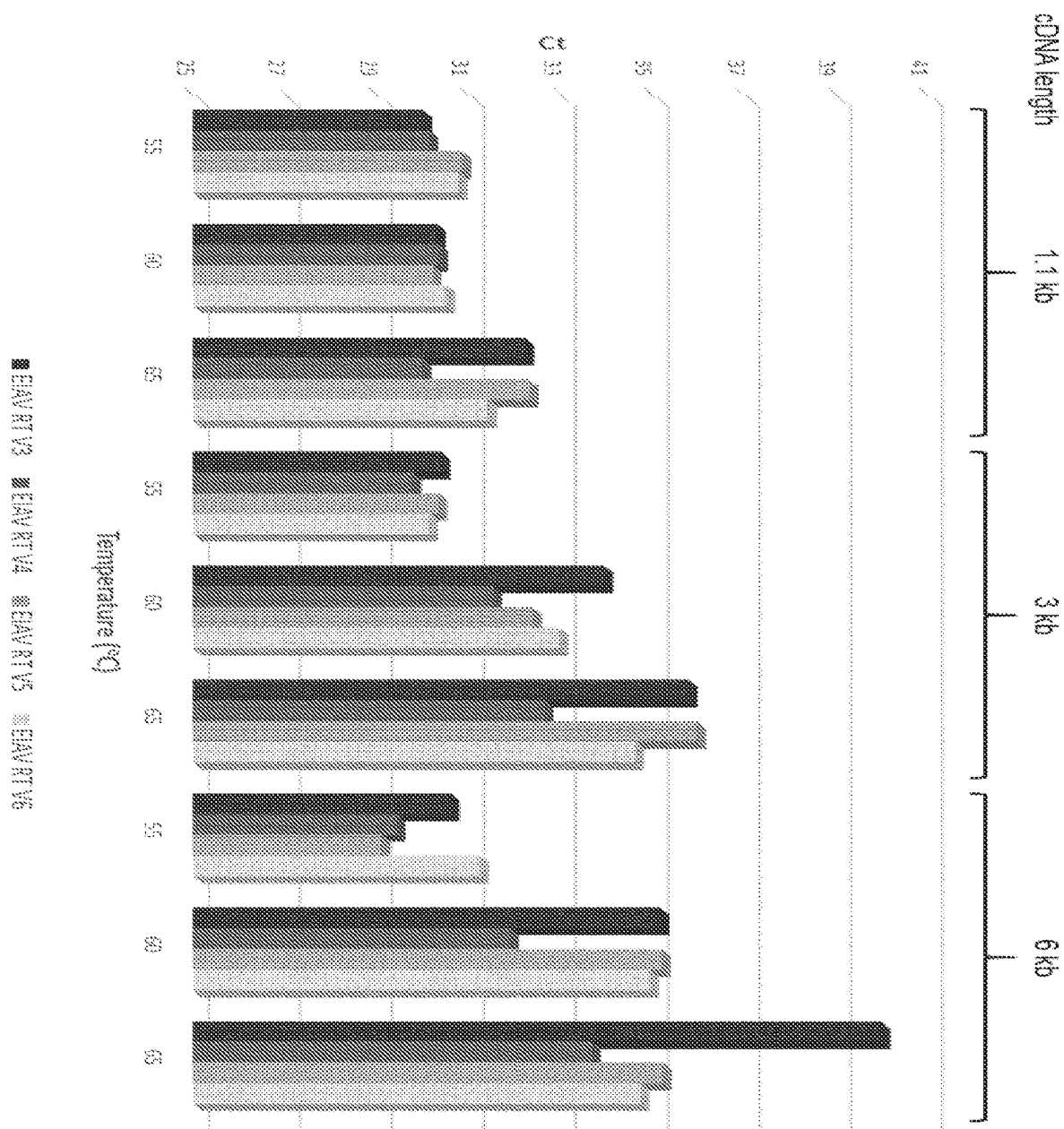


Fig. 6

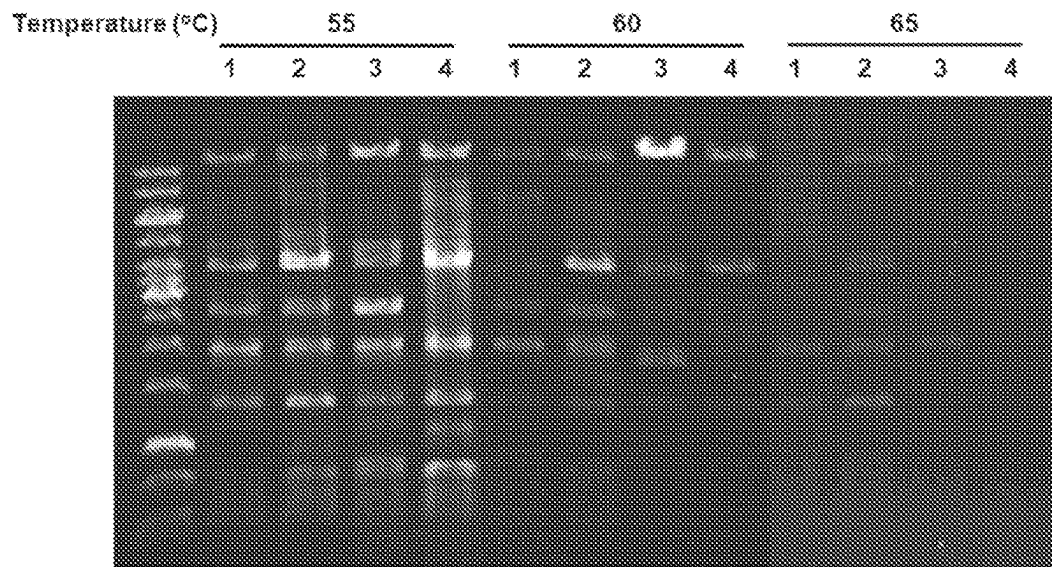


Fig. 7

1

THERMOSTABLE VIRAL REVERSE TRANSCRIPTASE

This application includes a Substitute Computer Readable Form (CFR) copy of a sequence listing, which is hereby incorporated by reference. The Substitute Computer Readable Form (CFR) copy of the sequence listing was submitted on Jul. 20, 2022 via EFS-web as an ASCII text file named "QIA-52303-Substitute-Sequence-Listing.txt". The ASCII text file was created on Jul. 20, 2022, and is 72 KB in size.

FIELD OF THE INVENTION

The present invention provides novel engineered reverse transcriptase enzymes for improved detection and analysis of RNA. The present invention also provides methods for amplifying template nucleic acids using such reverse transcriptase enzymes and methods of producing said enzymes. This invention addresses deficiencies of current state of the art reverse transcriptase enzymes in RNA detection and analysis including deficiencies in detection sensitivity, specificity, side enzyme activities, enzyme stability and synthesis capacity, especially when using template nucleic acids ranging in length, secondary structure and nucleotide content.

TECHNOLOGICAL BACKGROUND

Detection and analysis of RNA can provide insight into important biological phenomena like gene expression, gene silencing, the presence and absence of important pathogens, particularly RNA viruses, and other medically important states. The ability to monitor presence or absence or changes in the amounts of messenger RNA, noncoding RNA and viral RNA allows diagnosis of important disease states like cancer or viral infections and generally facilitates research into biological processes. For these reasons, improvements in the state of the art in these analyses are highly sought.

The most common methods of RNA analysis depend on reverse transcriptase enzymes to catalyze the synthesis of complementary DNA (cDNA) using an RNA template (Spiegelman 1971, Telesnitsky 1997), which is a critical step in several related analytic and preparative methods. For example, reverse transcription PCR (RT-PCR) (Lee 1989) and its variants quantitative RT-PCR (RT qPCR), realtime quantitative RT-PCR (RT RT-PCR) (Owarzek 1992, Paria 1993, Gibson 1996) and digital RT-PCR (RT dPCR) (Warren 2006, Sanders 2018) are fundamentally two-step processes in which cDNA is synthesized by reverse transcription and amplified by PCR. Typically, these two functions are provided by separate enzymes, a reverse transcriptase (RT), often a Moloney murine leukemia virus (MMLV) RT derivative and a thermostable DNA polymerase (Pol), typically Taq Pol. The performance of these methods depends on both activities, but improvements are most commonly sought by focusing on the RT enzyme since the PCR step is generally better established and more robust.

In addition to RT-PCR, areas of RNA-centered genomics focused on transcripts, noncoding RNA, and viral genomics and metagenomics rely on variants of analytic approaches referred to collectively as RNA-Seq (reviewed in Hrdlickova 2017, Boone et al 2018). Different RNA-Seq strategies are used to address different biological questions associated with phenomena like variations in gene expression and the relative abundance of transcripts, diversity of viral genomic sequences and their evolution, intron processing and splicing and other related phenomena. In all cases,

2

the enzyme component is critical to the generation of libraries used for sequence analysis and has a substantial impact on the overall quality and reliability of the analysis. Besides RT-PCR and RNA-Seq, cDNA synthesis has a more traditional use as a preparative method related to cloning cDNA for more focused analysis on transcripts and translated proteins (Spiegelman 1971, Burrell 1996). Optimal performance in each of these analyses places unique demands on the enzymes and the choice of RT has a strong impact on the effectiveness and reliability of the method (Okello 2010, Bustin 2015).

Engineering has refined the performance of RT in specific applications. Many RTs have inherent RNase H activities that digest the RNA component of RNA/DNA hybrids (Gerard 1975). This is a positive attribute for certain applications, but interferes with other common uses (Garces 1991). There is a secondary effect of disabling RNase H activity: besides eliminating its canonical function as a nuclease, eliminating RNase H activity by mutagenesis also improves thermostability. This allows synthesis at higher temperatures, thereby facilitating analysis of highly structured RNA targets. Additional research on MMLV RT indicates that certain domains, particularly the minor groove binding track (MGBT) and connection domain (CD), play roles in interacting with template (Beard 1994). Modifying either RNase H activity, the MGBT or CD or some combination can increase template switching (Garces 1991, Svarovskaia 2000) and affinity for template and, by extension, length of product. These improvements are beneficial to varying degrees in RT-PCR, preparation of RNA-Seq libraries and cDNA cloning.

Numerous native and engineered RTs have been examined to address the specialized needs of these different applications. Most common are derivatives of Moloney murine leukemia virus (MMLV) RT (Gerard 1975) or, less commonly, avian myeloblastosis (AMV) RT (Kacian 1971). Other RT enzymes have been derived from alternative retroviral enzymes (e.g., U.S. Pat. No. 7,560,117), phages (Moser 2012) and bacteria (Grabco 1996) but none fully addresses the needs for highly sensitive, highly specific, robust cDNA synthesis of targets ranging in length, secondary structure and nucleotide content. These limitations in the state of the art thus create a continuing need for improvements.

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Warren L, Bryder D, Weissman I L, Quake S R. Transcription factor profiling in individual hematopoietic progenitors by digital RT-PCR. *Proc Natl Acad Sci USA.* 2006 103(47): 17807-12.

SUMMARY OF THE INVENTION

This invention provides novel engineered reverse transcriptase enzymes that afford beneficial improvements in thermal stability, length of cDNAs synthesized using an RNA template, and elimination of secondary enzymatic activity. These improvements were incorporated into the parental equine infectious anemia virus reverse transcriptase (EIAV RT) molecule by directed mutagenesis of amino acid residues residing in thumb, connection, and RNase H domains. These mutations collectively eliminate RNase H activity and increase RT activity at elevated temperature and were identified from both rational design and random mutagenesis/screening. Combining these mutations allows significant improvement in performance including the capability to synthesize cDNAs of greater than 12 kb in length at the temperatures up to 65° C.

In a first aspect, the present invention relates to a reverse transcriptase (RT) comprising two subunits, wherein the two subunits are each encoded by a variant of the polynucleotide sequence of SEQ ID NO:1, wherein the amino acid sequence encoded by the variant is at least 90% identical to the amino acid sequence of SEQ ID NO:2 or a fragment thereof, and wherein the variant comprises one or more mutations in the polynucleotide sequence of SEQ ID NO:1 causing one or more amino acid exchanges relative to the amino acid sequence SEQ ID NO:2 in the minor groove binding track (MGBT) of the RT's thumb domain and/or the RT's connection domain (CD).

The one or more amino acid exchanges according to this aspect of the invention can be caused within the encoded amino acid sequence at the amino acid positions that correspond to positions 263, 265, 267, 269, and 271 of SEQ ID NO:2, wherein the amino acid exchanges are M263V, N265K, T267N, M269A, and S271Q.

Additionally or alternatively, the one or more amino acid exchanges can be caused within the encoded amino acid sequence at the amino acid positions that correspond to positions 356, 360, 362, and 363 of SEQ ID NO:2, wherein the amino acid exchanges are I356G or I356D or I356N, N360A, G362T or G362N, and W363K or W363N or W363I, preferably I356G, N360A, G362T and W363K.

According to one embodiment, the variant of SEQ ID NO:1 further comprises one or more mutations in the polynucleotide sequence of SEQ ID NO:1 causing one or more amino acid exchanges relative to SEQ ID NO:2 in the RT's RNase H domain. The one or more amino acid exchanges according to this aspect of the invention can be caused within the encoded amino acid sequence at the amino acid positions that correspond to positions 443, 470, 476, 491, 526 and 553 of SEQ ID NO:2, wherein the amino acid exchanges are D443G, V470F, E476Q, Q491R, R526H and K553R. In a preferred embodiment, the amino acid

exchanges are D443G and/or E476Q. In a highly preferred embodiment, the amino acid exchange is D443G.

According to another embodiment, the one or more mutations in the polynucleotide sequence of SEQ ID NO:1 cause one or more amino acid exchanges within the encoded amino acid sequence at the amino acid positions that correspond to positions 263, 265, 267, 269, 271 and 443 of SEQ ID NO:2, wherein the amino acid exchanges are M263V, N265K, T267N, M269A, S271Q and D443G. In a preferred embodiment, mutations in the polynucleotide sequence of SEQ ID NO:1 cause amino acid exchanges within the encoded amino acid sequence at all the amino acid positions that correspond to positions 263, 265, 267, 269, 271 and 443 of SEQ ID NO:2, wherein the amino acid exchanges are M263V, N265K, T267N, M269A, S271Q and D443G.

According to a further embodiment, the one or more mutations in the polynucleotide sequence of SEQ ID NO:1 cause one or more amino acid exchanges within the encoded amino acid sequence at the amino acid positions that correspond to positions 263, 265, 267, 269, 271, 356, 360, 362, 363 and 443 of SEQ ID NO:2, wherein the amino acid exchanges are M263V, N265K, T267N, M269A, S271Q, I356G, N360A, G362T, W363K and D443G. In a preferred embodiment, mutations in the polynucleotide sequence of SEQ ID NO:1 cause amino acid exchanges within the encoded amino acid sequence at all the amino acid positions that correspond to positions 263, 265, 267, 269, 271, 356, 360, 362, 363 and 443 of SEQ ID NO:2, wherein the amino acid exchanges are M263V, N265K, T267N, M269A, S271Q, I356G, N360A, G362T, W363K and D443G. Alternatively, mutations in the polynucleotide sequence of SEQ ID NO:1 cause amino acid exchanges within the encoded amino acid sequence at all the amino acid positions that correspond to positions 263, 265, 267, 269, 271, 356, 360, 362, 363 and 443 of SEQ ID NO:2, wherein the amino acid exchanges are M263V, N265K, T267N, M269A, S271Q, I356G, N360A, G362T, W363K and D443G. Yet alternatively, mutations in the polynucleotide sequence of SEQ ID NO:1 cause amino acid exchanges within the encoded amino acid sequence at all the amino acid positions that correspond to positions 263, 265, 267, 269, 271, 356, 360, 362, 363 and 443 of SEQ ID NO:2, wherein the amino acid exchanges are M263V, N265K, T267N, M269A, S271Q, I356D, N360A, G362T, W363K and D443G. Yet alternatively, mutations in the polynucleotide sequence of SEQ ID NO:1 cause amino acid exchanges within the encoded amino acid sequence at all the amino acid positions that correspond to positions 263, 265, 267, 269, 271, 356, 360, 362, 363 and 443 of SEQ ID NO:2, wherein the amino acid exchanges are M263V, N265K, T267N, M269A, S271Q, I356N, N360A, G362N, W363I and D443G.

According to a further embodiment, the one or more mutations in the polynucleotide sequence of SEQ ID NO:1 cause one or more amino acid exchanges within the encoded amino acid sequence at the amino acid positions that correspond to positions 263, 265, 267, 269, 271, 491, 526 and 443 of SEQ ID NO:2, wherein the amino acid exchanges are M263V, N265K, T267N, M269A, S271Q, Q491R, R526H and D443G. In a preferred embodiment, mutations in the polynucleotide sequence of SEQ ID NO:1 cause amino acid exchanges within the encoded amino acid sequence at all the amino acid positions that correspond to positions 263, 265, 267, 269, 271, 491, 526 and 443 of SEQ ID NO:2, wherein the amino acid exchanges are M263V, N265K, T267N, M269A, S271Q, Q491R, R526H and D443G. Alternatively, mutations in the polynucleotide sequence of SEQ ID NO:1 cause amino acid exchanges within the encoded amino acid

sequence at all the amino acid positions that correspond to positions 263, 265, 267, 269, 271, 491, 526 and 443 of SEQ ID NO:2, wherein the amino acid exchanges are M263V, N265K, T267N, M269A, S271Q, Q491R, K553R and D443G. Yet alternatively, mutations in the polynucleotide sequence of SEQ ID NO:1 cause amino acid exchanges within the encoded amino acid sequence at all the amino acid positions that correspond to positions 263, 265, 267, 269, 271, 491, 526 and 443 of SEQ ID NO:2, wherein the amino acid exchanges are M263V, N265K, T267N, M269A, S271Q, Q491R, R526H, K553R and D443G.

According to a further embodiment, the one or more mutations in the polynucleotide sequence of SEQ ID NO:1 cause one or more amino acid exchanges within the encoded amino acid sequence at the amino acid positions that correspond to positions 263, 265, 267, 269, 271, 356, 360, 362, 363, 491, 526 and 443 of SEQ ID NO:2, wherein the amino acid exchanges are M263V, N265K, T267N, M269A, S271Q, I356G, N360A, G362T, W363K, Q491R, R526H and D443G. In a preferred embodiment, mutations in the polynucleotide sequence of SEQ ID NO:1 cause amino acid exchanges within the encoded amino acid sequence at all the amino acid positions that correspond to positions 263, 265, 267, 269, 271, 356, 360, 362, 363, 491, 526 and 443 of SEQ ID NO:2, wherein the amino acid exchanges are M263V, N265K, T267N, M269A, S271Q, I356G, N360A, G362T, W363K, Q491R, R526H and D443G.

According to a preferred embodiment, a first subunit of the two subunits comprises the complete amino acid sequence encoded by the variant of SEQ ID NO:1 and a second subunit of the two subunits comprises a proteolytic fragment of the complete amino acid sequence encoded by the variant of SEQ ID NO:1.

In another aspect, the invention refers to a method for amplifying template nucleic acids comprising contacting the template nucleic acids with an RT of the invention. In one embodiment, the method is reverse transcription (RT) PCR.

In a further aspect, the present invention provides a kit comprising the RT of the invention and a buffer.

Another aspect of the invention relates to a polynucleotide encoding an RT of the invention. A further aspect of the invention relates to a vector comprising the polynucleotide according to the invention. In another aspect, the invention relates to transformed host cells comprising said vector.

In another aspect, the invention provides an RT obtainable by expression of the polynucleotide or the vector according to the invention in a host cell. In a preferred embodiment, the host cell is *E. coli*.

In yet another embodiment, the invention relates to a method of producing an RT of the invention comprising isolating the RT from the transformed host cell.

BRIEF DESCRIPTION OF THE DRAWINGS

The present disclosure will be more readily appreciated by reference to the following detailed description when being considered in connection with the accompanying drawings in which:

FIG. 1 shows the results of a thermal stability assay using different RTs that have been subjected to a preincubation step at the indicated temperatures prior to the RT reaction at 42° C.

FIG. 2 depicts the results of a thermal stability assay measuring cDNA synthesis at elevated temperature using different RTs. Panel (a) shows the results of cDNA synthesis reactions at 50° C. Panel (b) shows the results of 2-step

RT-PCR reactions where cDNA was generated at 50° C. (upper panel) and 60° C. (lower panel).

FIG. 3 shows the effect of temperature on cDNA yields in a one step RT-qPCR assay using a state of the art MMLV RT with reduced RNase H activity (EnzScript) and RNase H^{neg} EIAV RT variants according to the invention, wherein EIAV RT V3 additionally comprises point mutations within MGBT.

FIG. 4 shows the effect of temperature on the yield of cDNAs of different lengths in a 2-step RT-qPCR assay using a state of the art MMLV RT (SuperScript IV) and EIAV RT variants according to the invention.

FIG. 5 shows a time course of 9 kb cDNA synthesis at 55° C. by EIAV RT V3 and the state of the art MMLV RT SuperScript IV.

FIG. 6 shows the effect of temperature on cDNA yields in a 2-step RT-qPCR assay using different EIAV RT variants according to the invention.

FIG. 7 shows the effect of temperature on cDNA yields in a long-range endpoint PCR assay using different EIAV RT variants according to the invention.

DETAILED DESCRIPTION OF THE INVENTION

Definitions

Unless defined otherwise, all technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art (e.g., in cell culture, molecular genetics, nucleic acid chemistry, hybridization techniques and biochemistry).

In practicing the present invention, many conventional techniques in molecular biology, microbiology, and recombinant DNA may be used. These techniques are well known and are explained in, for example, Current Protocols in Molecular Biology, Volumes I, II, and III, 1997 (F. M. Ausubel ed.); Sambrook et al., 1989, Molecular Cloning: A Laboratory Manual, Second Edition, Cold Spring Harbor Laboratory Press, Cold Spring Harbor, N.Y.; DNA Cloning: A Practical Approach, Volumes I and II, 1985 (D. N. Glover ed.); Oligonucleotide Synthesis, 1984 (M. L. Gait ed.); Nucleic Acid Hybridization, 1985 (Hames and Higgins); Transcription and Translation, 1984 (Hames and Higgins eds.); Animal Cell Culture, 1986 (R. I. Freshney ed.); Immobilized Cells and Enzymes, 1986 (IRL Press); Perbal, 1984, A Practical Guide to Molecular Cloning; the series, Methods In Enzymology (Academic Press, Inc.); Gene Transfer Vectors for Mammalian Cells, 1987 (J. H. Miller and M. P. Calos eds., Cold Spring Harbor Laboratory); and Methods in Enzymology Vol. 154 and Vol. 155 (Wu and Grossman, and Wu, eds., respectively).

As used herein, the term “comprising” is to be construed as encompassing both “including” and “consisting of”, both meanings being specifically intended, and hence individually disclosed embodiments in accordance with the present invention.

The term “nucleic acid sequence”, “nucleotide sequence” or “polynucleotide sequence” as used herein refers to an oligonucleotide, nucleotide or polynucleotide and fragments and portions thereof and to DNA or RNA of genomic or synthetic origin, which may be single or double stranded and represent the sense or antisense strand. The sequence may be a non-coding sequence, a coding sequence or a mixture of both.

A “variant” of a polynucleotide sequence (e.g., RNA or DNA) comprises one or more mutations within the poly-

nucleotide sequence relative to another polynucleotide sequence wherein one or more nucleic acid residues are inserted into, deleted from and/or substituted into the nucleic acid sequence. Said one or more mutations can cause one or more amino acid exchanges within the amino acid sequence the variant encodes for as compared to another amino acid sequence (i.e. a “non-silent mutation”). Variants also include nucleic acid sequences wherein one or more codons have been replaced by their synonyms which does not cause an amino acid exchange and is thus called a “silent mutation”.

The term “identity” refers to a relationship between the sequences of two or more polypeptide molecules or two or more nucleic acid molecules, as determined by aligning and comparing the sequences. “Percent identity” means the percent of identical residues between the amino acids or nucleotides in the compared molecules and is calculated based on the size of the smallest of the molecules being compared. For these calculations, gaps in alignments (if any) are preferably addressed by a particular mathematical model or computer program (i.e., an “algorithm”). Methods that can be used to calculate the identity of the aligned nucleic acids or polypeptides include those described in Computational Molecular Biology, (Lesk, A. M., ed.), 1988, New York: Oxford University Press; Biocomputing Informatics and Genome Projects, (Smith, D. W., ed.), 1993, New York: Academic Press; Computer Analysis of Sequence Data, Part I, (Griffin, A. M., and Griffin, H. G., eds.), 1994, New Jersey: Humana Press; von Heinje, G., 1987, Sequence Analysis in Molecular Biology, New York: Academic Press; Sequence Analysis Primer, (Gribskov, M. and Devereux, J., eds.), 1991, New York: M. Stockton Press; and Carillo et al, 1988, SI AM J. Applied Math. 48: 1073.

In calculating percent identity, the sequences being compared are typically aligned in a way that gives the largest match between the sequences. Computer programs that can be used to determine percent identity are discussed, e.g., by Pearson (Pearson, 2013, Curr. Protoc. Bioinform. 42:3.1.1-3.1.8). Such computer algorithms are used to align the two polypeptides or polynucleotides for which the percent sequence identity is to be determined. The sequences are aligned for optimal matching of their respective amino acid or nucleotide (the “matched span”, as determined by the algorithm).

Certain alignment schemes for aligning two sequences may result in matching of only a short region of the two sequences, and this small aligned region may have very high sequence identity even though there is no significant relationship between the two full-length sequences. Accordingly, the selected alignment method can be adjusted if so desired to result in an alignment that spans at least 50 or other number of contiguous nucleotides or amino acids.

The nucleic acid sequences of the present invention can be prepared using standard techniques well known to one of skill in the art. The term “encoding” or “coding” refers to the inherent property of specific sequences of nucleotides in a nucleic acid, such as a gene in chromosome or an mRNA, to serve as templates for synthesis of other polymers and macromolecules in biological processes having a defined sequence of nucleotides (i.e., rRNA, tRNA, other RNA molecules) or amino acids and the biological properties resulting therefrom. Thus a gene encodes a protein, if transcription and translation of mRNA produced by that gene produces the protein in a cell or other biological system. Both the coding strand, the nucleotide sequence of which is identical to the mRNA sequence and is usually provided in sequence listings, and non-coding strand, used as the template for the transcription, of a gene or cDNA can

be referred to as encoding the protein or other product of that gene or cDNA. A nucleic acid that encodes a protein includes any nucleic acids that have different nucleotide sequences but encode the same amino acid sequence of the protein due to the degeneracy of the genetic code. Nucleic acids and nucleotide sequences that encode proteins may include introns.

The term "polypeptide" is used interchangeably with "amino acid sequence" or "protein" and refers to polymers of amino acids of any length. These terms also include proteins that are post-translationally modified through reactions that include, but are not limited to, glycosylation, acetylation, phosphorylation or protein processing. Modifications and changes, for example fusions to other proteins, amino acid sequence substitutions, deletions or insertions, can be made in the structure of a polypeptide while the molecule maintains its biological functional activity. For example certain amino acid sequence substitutions can be made in a polypeptide or its underlying nucleic acid coding sequence and a protein can be obtained with like properties. Amino acid modifications can be prepared for example by performing site-specific mutagenesis or polymerase chain reaction mediated mutagenesis on its underlying nucleic acid sequence.

The term "expressed" as used herein refers to transcription and/or translation of a heterologous nucleic acid sequence within a host cell. "Production" would involve both transcription and translation. The level of expression in a host cell may be determined on the basis of either the amount of corresponding mRNA that is present in the cell, and production would be based on the amount of the desired polypeptide. For example, mRNA transcribed from a selected sequence can be quantitated by Northern blot hybridization, ribonuclease RNA protection, in situ hybridization to cellular RNA or by PCR (see Sambrook et al. (1989), supra; Ausubel et al. (1994 updated), supra). Proteins encoded by a selected sequence can be quantitated by various methods, e.g., by ELISA, by Western blotting, by radioimmunoassays, by immunoprecipitation, by assaying for the biological activity of the protein, or by immunostaining of the protein followed by FACS analysis PCR (see Sambrook et al. (1989), supra; Ausubel et al. (1994 updated), supra).

The term "PCR" refers to polymerase chain reaction, which is a standard method in molecular biology for DNA amplification.

"RT-PCR" relates to reverse transcription polymerase chain reaction, a variant of PCR commonly used for the detection and quantification of RNA. RT-PCR comprises two steps, synthesis of complementary DNA (cDNA) from RNA by reverse transcription and amplification of the generated cDNA by PCR. Variants of RT-PCR include quantitative RT-PCR (RT-qPCR), real-time RT-PCR, digital RT-PCR (dRT-PCR) or digital droplet RT-PCR (ddRT-PCR).

In the following, the invention will be explained in more detail with reference to the accompanying figures. It will be obvious for a person skilled in the art that these embodiments and items only depict examples of a plurality of possibilities. Hence, the embodiments shown here should not be understood to form a limitation of these features and configurations. Any possible combination and configuration of the described features can be chosen according to the scope of the invention.

Polymerases/Enzymes

Equine infectious anemia virus reverse transcriptases (EIAV RTs) are active in vivo as a heterodimer comprising subunits of 66 kDa and 51 kDa derived from a single open

reading frame. The smaller subunit of 51 kDa is a proteolytic fragment of the bigger 66 kDa subunit. Expression of the open reading frame of SEQ ID NO:1 in *E. coli* gives rise to both the 66 kDa and the 51 kDa subunits that can form homodimers (i.e., p66/p66 and p51/p51) and heterodimers (i.e., p66/p51). Some commercial preparations of RTs are mixtures of the homodimers and heterodimers. While both types of dimers are active in DNA synthesis, the heterodimer presents with higher stability and processivity than the homodimers.

A preferred embodiment of the RT according to the invention thus is an enzyme preparation being enriched for heterodimers. Enrichment can be done through any method known to the skilled person, for example chromatography including Heparin, hydrophobic interaction chromatography (HIC), and ion exchanges. An enzyme preparation is enriched for heterodimers if the content of heterodimers is at least 25% higher as compared to the pre-enrichment content of heterodimers in the enzyme preparation. In some embodiments of this aspect of the invention, the content of heterodimers is enriched by at least 30% as compared to the pre-enrichment content of heterodimers in the enzyme preparation. In another embodiment, the content of heterodimers is enriched by at least 50% as compared to the pre-enrichment content of heterodimers in the enzyme preparation. In a preferred embodiment of this aspect of the invention, the content of heterodimers is enriched by at least 75% as compared to the pre-enrichment content of heterodimers in the enzyme preparation. In a more preferred embodiment of this aspect of the invention, the content of heterodimers is enriched by at least 100% as compared to the pre-enrichment content of heterodimers in the enzyme preparation. In a particularly preferred embodiment of this aspect of the invention, the heterodimer is purified to apparent homogeneity.

In a first aspect, the present invention provides a reverse transcriptase (RT) comprising two subunits, wherein the two subunits are each encoded by a variant of the polynucleotide sequence of SEQ ID NO:1, wherein the amino acid sequence encoded by the variant is at least 90% identical to the amino acid sequence of SEQ ID NO:2 or a fragment thereof, and wherein the variant comprises one or more mutations in the polynucleotide sequence of SEQ ID NO:1 causing one or more amino acid exchanges relative to the amino acid sequence of SEQ ID NO:2 in the minor groove binding track (MGBT) of the RT's thumb domain and/or the RT's connection domain (CD).

The variant of SEQ ID NO:1 can comprise an arbitrary number of codons that have been replaced by their synonyms (see Table 1) which does not cause an amino acid exchange within the amino acid sequence relative to SEQ ID NO:2. Additionally, the variant comprises at least one mutation that causes an amino acid exchange relative to the amino acid sequence of SEQ ID NO:2. In some embodiments, the variant comprises at least two mutations that cause amino acid exchanges relative to the amino acid sequence of SEQ ID NO:2. In another embodiment, the variant comprises at least three mutations that cause amino acid exchanges relative to the amino acid sequence of SEQ ID NO:2. In another embodiment, the variant comprises at least four mutations that cause amino acid exchanges relative to the amino acid sequence of SEQ ID NO:2. In another embodiment, the variant comprises at least five mutations that cause amino acid exchanges relative to the amino acid sequence of SEQ ID NO:2. In another embodiment, the variant comprises at least six mutations that cause amino acid exchanges relative to the amino acid sequence of SEQ

11

ID NO:2. In another embodiment, the variant comprises at least eight mutations that cause amino acid exchanges relative to the amino acid sequence of SEQ ID NO:2. In another embodiment, the variant comprises at least ten mutations that cause amino acid exchanges relative to the amino acid sequence of SEQ ID NO:2.

TABLE 1

Genetic code		
#	Amino acid	codon
	Start	AUG
1	Met	AUG
1	Trp	UGG
1	Sec	(UGA)
1	Pyl	(UAG)
2	Tyr	UAU
2	Phe	UUU
2	Cys	UGU
2	Asn	AAU
2	Asp	GAU
2	Gln	CAA
2	Glu	GAA
2	His	CAU CAC
2	Lys	AAA AAG
3	Ile	AUU AUC AUA
4	Gly	GGU GGC GGA GGG
4	Ala	GCU GCC GCA GCG
4	Val	GUU GUC GUA GUG
4	Thr	ACU ACC ACA ACG
4	Pro	CCU CCC CCA CCG
6	Leu	CUU CUC CUA CUG UUA
6	Ser	UCU UCC UCA UCG AGU
6	Arg	CGU CGC CGA CGG AGA
3	Stop	UAA UAG UGA

In some embodiments according to this aspect of the invention, the variant of SEQ ID NO:1 comprises one mutation that causes an amino acid exchange relative to the amino acid sequence of SEQ ID NO:2. In another embodiment, the variant comprises two mutations that cause amino acid exchanges relative to the amino acid sequence of SEQ ID NO:2. In another embodiment, the variant comprises three mutations that cause amino acid exchanges relative to the amino acid sequence of SEQ ID NO:2. In another embodiment, the variant comprises four mutations that cause amino acid exchanges relative to the amino acid sequence of SEQ ID NO:2. In another embodiment, the variant comprises five mutations that cause amino acid exchanges relative to the amino acid sequence of SEQ ID NO:2. In another embodiment, the variant comprises six

12

mutations that cause amino acid exchanges relative to the amino acid sequence of SEQ ID NO:2. In another embodiment, the variant comprises eight mutations that cause amino acid exchanges relative to the amino acid sequence of SEQ ID NO:2. In another embodiment, the variant comprises ten mutations that cause amino acid exchanges relative to the amino acid sequence of SEQ ID NO:2. In another embodiment, the variant comprises twelve mutations that cause amino acid exchanges relative to the amino acid sequence of SEQ ID NO:2.

The one or more amino acid exchanges relative to the amino acid sequence of SEQ ID NO:2 can locate to the minor groove binding track (MGBT) of the RT's thumb domain spanning amino acid residues 255-298 of SEQ ID NO:2, the RT's connection domain (CD) spanning amino acid residues 320-414 of SEQ ID NO:2, and/or the RT's RNase H domain spanning amino acid residues 438-553 of SEQ ID NO:2. In some embodiments according to this aspect of the invention, the at least one amino acid exchange relative to the amino acid sequence of SEQ ID NO:2 locates to the MGBT of the RT's thumb domain. In other embodiments according to this aspect of the invention, the at least one amino acid exchange relative to the amino acid sequence of SEQ ID NO:2 locates to the RT's CD. In further embodiments according to this aspect of the invention, the at least one amino acid exchange relative to the amino acid sequence of SEQ ID NO:2 locates to the RT's RNase H domain. In preferred embodiments, the at least two amino acid exchanges relative to the amino acid sequence of SEQ ID NO:2 locate to the MGBT of the RT's thumb domain and the RT's CD. In another preferred embodiments, the at least two amino acid exchanges relative to the amino acid sequence of SEQ ID NO:2 locate to the MGBT of the RT's thumb domain and the RT's RNase H domain. In particularly preferred embodiments, the at least three amino acid exchanges relative to the amino acid sequence of SEQ ID NO:2 locate to the MGBT of the RT's thumb domain, the RT's RNase H domain and the RT's CD.

In some embodiments according to this aspect of the invention, the variant of SEQ ID NO:1 comprises one or more mutations that cause one amino acid exchange relative to the amino acid sequence of SEQ ID NO:2. This one amino acid exchange relative to the amino acid sequence of SEQ ID NO:2 can locate to the minor groove binding track (MGBT) of the RT's thumb domain spanning amino acid residues 255-298 of SEQ ID NO:2, the RT's connection domain (CD) spanning amino acid residues 320-414 of SEQ ID NO:2, and/or the RT's RNase H domain spanning amino acid residues 438-553 of SEQ ID NO:2. In certain embodiments, this one amino acid exchange relative to the amino acid sequence of SEQ ID NO:2 locates to the MGBT of the RT's thumb domain. In other embodiments, this one amino acid exchange relative to the amino acid sequence of SEQ ID NO:2 locates to the RT's CD. In further embodiments, this one amino acid exchange relative to the amino acid sequence of SEQ ID NO:2 locates to the RT's RNase H domain. In preferred embodiments, the variant comprises mutations that cause two amino acid exchanges relative to the amino acid sequence of SEQ ID NO:2. These two amino acid exchanges relative to the amino acid sequence of SEQ ID NO:2 locate to the MGBT of the RT's thumb domain and the RT's CD. In another preferred embodiments, these two amino acid exchanges relative to the amino acid sequence of SEQ ID NO:2 locate to the MGBT of the RT's thumb domain and the RT's RNase H domain. In particularly preferred embodiments, the variant comprises mutations that cause three amino acid exchanges relative to the amino acid

sequence of SEQ ID NO:2. These three amino acid exchanges relative to the amino acid sequence of SEQ ID NO:2 locate to the MGBT of the RT's thumb domain, the RT's RNase H domain and the RT's CD.

In some embodiments according to this aspect of the invention, the amino acid sequence encoded by the variant of SEQ ID NO:1 is at least 90% identical to the amino acid sequence of SEQ ID NO:2 or a fragment thereof. In other embodiments, the amino acid sequence encoded by the variant is at least 95% identical to the amino acid sequence of SEQ ID NO:2 or a fragment thereof. In preferred embodiments according to this aspect of the invention, the amino acid sequence encoded by the variant is at least 97% identical to the amino acid sequence of SEQ ID NO:2 or a fragment thereof. An amino acid sequence that is at least 90%, at least 95% or at least 97% identical to the amino acid sequence of SEQ ID NO:2 or a fragment thereof retains the biological function of the amino acid sequence of SEQ ID NO:2 or a fragment thereof. More specifically, an amino acid sequence that is at least 90%, at least 95% or at least 97% identical to the amino acid sequence of SEQ ID NO:2 or a fragment thereof retains the reverse transcriptase activity of EIAV-RT.

According to one embodiment, the one or more amino acid exchanges according to this aspect of the invention can be caused within the encoded amino acid sequence at the amino acid positions that correspond to positions 263, 265, 267, 269, and 271 of SEQ ID NO:2, wherein the amino acid exchanges are M263V, N265K, T267N, M269A, and S271Q. Preferably, the EIAV RT comprises at least two of said amino acid exchanges within its MGBT. More preferably, the EIAV RT comprises at least three of said amino acid exchanges within its MGBT. In some further embodiments, the EIAV RT comprises two of said amino acid exchanges within its MGBT. More preferably, the EIAV RT comprises three of said amino acid exchanges within its MGBT. Particularly preferred, the EIAV RT comprises all of said amino acid exchanges within its MGBT.

In addition or alternatively to the at least one amino acid exchange within the RT's MGBT, the one or more amino acid exchanges can be caused within the encoded amino acid sequence at the amino acid positions that correspond to positions 356, 360, 362, and 363 of SEQ ID NO:2, wherein the amino acid exchanges are I356G or I356D or I356N, N360A, G362T or G362N, and W363K or W363N or W363L, preferably I356G, N360A, G362T and W363K. Preferably, the EIAV RT comprises at least two of said amino acid exchanges within its CD. More preferably, the EIAV RT comprises at least three of said amino acid exchanges within its CD. In some further embodiments, the EIAV RT comprises two of said amino acid exchanges within its CD. More preferably, the EIAV RT comprises three of said amino acid exchanges within its CD. Particularly preferred, the EIAV RT comprises all of said amino acid exchanges within its CD.

In addition or alternatively to the at least one amino acid exchange within the RT's MGBT, the one or more amino acid exchanges can be caused within the encoded amino acid sequence at the amino acid positions that correspond to positions 443, 470, 476, 491, 526 and 553 of SEQ ID NO:2, wherein the amino acid exchanges are D443G, V470F, E476Q, Q491R, R526H and K553R. According to one embodiment, the EIAV RT comprises an amino acid exchange corresponding to position D443G and/or E476Q of SEQ ID NO:2. According to one embodiment, the EIAV RT comprises one amino acid exchange corresponding to position D443G of SEQ ID NO:2. This EIAV RT has an

amino acid sequence of SEQ ID NO:4 that is encoded by the nucleic acid sequence of SEQ ID NO:3 or a synonymous variant thereof. According to another embodiment, the EIAV RT comprises two amino acid exchanges corresponding to positions D443G and E476Q of SEQ ID NO:2. Said EIAV RT has an amino acid sequence of SEQ ID NO:6 that is encoded by the nucleic acid sequence of SEQ ID NO:5 or a synonymous variant thereof. In a preferred embodiment, one of the amino acid exchanges is D443G. In other embodiments, the EIAV RT comprises at least two of the aforementioned amino acid exchanges within its RNase H domain. According to further embodiments, the EIAV RT comprises at least three of said amino acid exchanges within its RNase H domain. In some further embodiments, the EIAV RT comprises two of the aforementioned amino acid exchanges within its RNase H domain. According to further embodiments, the EIAV RT comprises three of said amino acid exchanges within its RNase H domain. Particularly preferred, the EIAV RT comprises the amino acid exchanges D443G, Q491R, and R526H.

In some embodiments according to this aspect of the invention, the one or more mutations in the polynucleotide sequence of SEQ ID NO:1 cause one or more amino acid exchanges within the encoded amino acid sequence at the amino acid positions that correspond to positions 263, 265, 267, 269, 271 and 443 of SEQ ID NO:2, wherein the amino acid exchanges are M263V, N265K, T267N, M269A, S271Q and D443G. Preferably, the EIAV RT comprises at least two of the aforementioned amino acid exchanges, wherein at least one of said amino acid exchanges locates to the RNase H domain and at least one of said amino acid exchanges locates to the MGBT. More preferably, the EIAV RT comprises at least three of the aforementioned amino acid exchanges, wherein at least one of said amino acid exchanges locates to the RNase H domain and at least two of said amino acid exchanges locate to the MGBT. According to another preferred embodiment, the EIAV RT comprises at least four of the aforementioned amino acid exchanges, wherein at least one of said amino acid exchanges locates to the RNase H domain and at least three of said amino acid exchanges locate to the MGBT. According to a further preferred embodiment, the EIAV RT comprises at least five of the aforementioned amino acid exchanges, wherein at least one of said amino acid exchanges locates to the RNase H domain and at least four of said amino acid exchanges locate to the MGBT. In some further embodiments, the EIAV RT comprises two of the aforementioned amino acid exchanges, wherein one of said amino acid exchanges locates to the RNase H domain and one of said amino acid exchanges locates to the MGBT. Preferably, the EIAV RT comprises three of the aforementioned amino acid exchanges, wherein one of said amino acid exchanges locates to the RNase H domain and two of said amino acid exchanges locate to the MGBT. According to another preferred embodiment, the EIAV RT comprises four of the aforementioned amino acid exchanges, wherein one of said amino acid exchanges locates to the RNase H domain and three of said amino acid exchanges locate to the MGBT. According to a further preferred embodiment, the EIAV RT comprises five of the aforementioned amino acid exchanges, wherein one of said amino acid exchanges locates to the RNase H domain and four of said amino acid exchanges locate to the MGBT. Particularly preferred, the EIAV RT comprises all of the aforementioned amino acid exchanges. The latter EIAV RT has an amino acid sequence of SEQ ID NO:8 that is encoded by the nucleic acid sequence of SEQ ID NO:7 or a synonymous variant thereof.

In further embodiments according to this aspect of the invention, the one or more mutations in the polynucleotide sequence of SEQ ID NO:1 cause one or more amino acid exchanges within the encoded amino acid sequence at the amino acid positions that correspond to positions 263, 265, 267, 269, 271, 356, 360, 362, 363 and 443 of SEQ ID NO:2, wherein the amino acid exchanges are M263V, N265K, T267N, M269A, S271Q, I356G, N360A, G362T, W363K and D443G. Preferably, the EIAV RT comprises at least three of the aforementioned amino acid exchanges, wherein at least one of said amino acid exchanges locates to the RNase H domain, at least one of said amino acid exchanges locates to the MGBT and at least one of said amino acid exchanges locates to the CD. According to another preferred embodiment, the EIAV RT comprises at least four of the aforementioned amino acid exchanges, wherein at least one of said amino acid exchanges locates to the RNase H domain, at least one of said amino acid exchanges locates to the MGBT and at least one of said amino acid exchanges locates to the CD. According to further preferred embodiments, the EIAV RT comprises at least five of the aforementioned amino acid exchanges, wherein at least one of said amino acid exchanges locates to the RNase H domain, at least two of said amino acid exchanges locate to the MGBT and at least two of said amino acid exchanges locate to the CD. According to other preferred embodiments, the EIAV RT comprises at least six of the aforementioned amino acid exchanges, wherein at least one of said amino acid exchanges locates to the RNase H domain, at least two of said amino acid exchanges locate to the MGBT and at least two of said amino acid exchanges locate to the CD. According to more preferred embodiments, the EIAV RT comprises at least seven of the aforementioned amino acid exchanges, wherein at least one of said amino acid exchanges locates to the RNase H domain, at least three of said amino acid exchanges locate to the MGBT and at least three of said amino acid exchanges locate to the CD. According to further more preferred embodiments, the EIAV RT comprises at least eight of the aforementioned amino acid exchanges, wherein at least one of said amino acid exchanges locates to the RNase H domain, at least three of said amino acid exchanges locate to the MGBT and at least three of said amino acid exchanges locate to the CD. According to other more preferred embodiments, the EIAV RT comprises at least nine of the aforementioned amino acid exchanges, wherein at least one of said amino acid exchanges locates to the RNase H domain, at least four of said amino acid exchanges locate to the MGBT and at least four of said amino acid exchanges locate to the CD. In some further embodiments, the EIAV RT comprises three of the aforementioned amino acid exchanges, wherein one of said amino acid exchanges locates to the RNase H domain, one of said amino acid exchanges locates to the MGBT and one of said amino acid exchanges locates to the CD. According to another embodiment, the EIAV RT comprises four of the aforementioned amino acid exchanges, wherein one of said amino acid exchanges locates to the RNase H domain and one or two of said amino acid exchanges locate to the MGBT and the CD respectively. According to further embodiments, the EIAV RT comprises five of the aforementioned amino acid exchanges, wherein one of said amino acid exchanges locates to the RNase H domain, two of said amino acid exchanges locate to the MGBT and two of said amino acid exchanges locate to the CD. According to other embodiments, the EIAV RT comprises six of the aforementioned amino acid exchanges, wherein one of said amino acid exchanges locates to the RNase H domain and two or

three of said amino acid exchanges locate to the MGBT and the CD respectively. According to more preferred embodiments, the EIAV RT comprises seven of the aforementioned amino acid exchanges, wherein one of said amino acid exchanges locates to the RNase H domain, three of said amino acid exchanges locate to the MGBT and three of said amino acid exchanges locate to the CD. According to further more preferred embodiments, the EIAV RT comprises eight of the aforementioned amino acid exchanges, wherein one of said amino acid exchanges locates to the RNase H domain and three or four of said amino acid exchanges locate to the MGBT and the CD respectively. According to other more preferred embodiments, the EIAV RT comprises nine of the aforementioned amino acid exchanges, wherein one of said amino acid exchanges locates to the RNase H domain, four of said amino acid exchanges locate to the MGBT and four of said amino acid exchanges locate to the CD. Particularly preferred, the EIAV RT comprises all of the aforementioned amino acid exchanges. The latter EIAV RT has an amino acid sequence of SEQ ID NO:10 that is encoded by the nucleic acid sequence of SEQ ID NO:9 or a synonymous variant thereof.

According to another embodiment, the mutations in the polynucleotide sequence of SEQ ID NO:1 cause amino acid exchanges within the encoded amino acid sequence at all the amino acid positions that correspond to positions 263, 265, 267, 269, 271, 356, 360, 362, 363 and 443 of SEQ ID NO:2, wherein the amino acid exchanges are M263V, N265K, T267N, M269A, S271Q, I356G, N360A, G362T, W363N and D443G.

According to another embodiment, the mutations in the polynucleotide sequence of SEQ ID NO:1 cause amino acid exchanges within the encoded amino acid sequence at all the amino acid positions that correspond to positions 263, 265, 267, 269, 271, 356, 360, 362, 363 and 443 of SEQ ID NO:2, wherein the amino acid exchanges are M263V, N265K, T267N, M269A, S271Q, I356G, N360A, G362T, W363N and D443G.

According to another embodiment, the mutations in the polynucleotide sequence of SEQ ID NO:1 cause amino acid exchanges within the encoded amino acid sequence at all the amino acid positions that correspond to positions 263, 265, 267, 269, 271, 356, 360, 362, 363 and 443 of SEQ ID NO:2, wherein the amino acid exchanges are M263V, N265K, T267N, M269A, S271Q, I356D, N360A, G362T, W363K and D443G.

According to another embodiment, the mutations in the polynucleotide sequence of SEQ ID NO:1 cause amino acid exchanges within the encoded amino acid sequence at all the amino acid positions that correspond to positions 263, 265, 267, 269, 271, 356, 360, 362, 363 and 443 of SEQ ID NO:2, wherein the amino acid exchanges are M263V, N265K, T267N, M269A, S271Q, I356N, N360A, G362N, W363I and D443G.

According to another embodiment, the mutations in the polynucleotide sequence of SEQ ID NO:1 cause amino acid exchanges within the encoded amino acid sequence at all the amino acid positions that correspond to positions 263, 265, 267, 269, 271, 443, 491 and 526 of SEQ ID NO:2, wherein the amino acid exchanges are M263V, N265K, T267N, M269A, S271Q, D443G, Q491R and R526H. Said EIAV RT has an amino acid sequence of SEQ ID NO:12 that is encoded by the nucleic acid sequence of SEQ ID NO:11 or a synonymous variant thereof.

According to another embodiment, the mutations in the polynucleotide sequence of SEQ ID NO:1 cause amino acid exchanges within the encoded amino acid sequence at all the

amino acid positions that correspond to positions 263, 265, 267, 269, 271, 443, 491 and 526 of SEQ ID NO:2, wherein the amino acid exchanges are M263V, N265K, T267N, M269A, S271Q, D443G, Q491R and K553R.

According to another embodiment, the mutations in the polynucleotide sequence of SEQ ID NO:1 cause amino acid exchanges within the encoded amino acid sequence at all the amino acid positions that correspond to positions 263, 265, 267, 269, 271, 443, 470, 491, 526 and 553 of SEQ ID NO:2, wherein the amino acid exchanges are M263V, N265K, T267N, M269A, S271Q, D443G, V470F, Q491R, R526H and K553R.

According to a further embodiment, the mutations in the polynucleotide sequence of SEQ ID NO:1 cause amino acid exchanges within the encoded amino acid sequence at all the amino acid positions that correspond to positions 263, 265, 267, 269, 271, 356, 360, 362, 363, 443, 491 and 526 of SEQ ID NO:2, wherein the amino acid exchanges are M263V, N265K, T267N, M269A, S271Q, I356G, N360A, G362T, W363K, D443G, Q491R and R526H. Said EIAV RT has an amino acid sequence of SEQ ID NO:14 that is encoded by the nucleic acid sequence of SEQ ID NO:13 or a synonymous variant thereof.

The following Table 2 summarizes the amino acid sequences of preferred embodiments according to the present invention and indicates one nucleic acid sequence encoding it. The skilled person knows that the amino acid sequences listed can also be encoded by nucleic acid sequences synonymous to those indicated in Table 2. The present invention also encompasses any of said synonymous nucleic acid sequences.

TABLE 2

Preferred embodiments and their nucleic/amino acid sequences		
Variant	Nucleic acid sequence	Amino acid sequence
EIAV RT WT	SEQ ID NO: 1	SEQ ID NO: 2
EIAV RT V1	SEQ ID NO: 3	SEQ ID NO: 4
EIAV RT V2	SEQ ID NO: 5	SEQ ID NO: 6
EIAV RT V3	SEQ ID NO: 7	SEQ ID NO: 8
EIAV RT V4	SEQ ID NO: 9	SEQ ID NO: 10
EIAV RT V5	SEQ ID NO: 11	SEQ ID NO: 12
EIAV RT V6	SEQ ID NO: 13	SEQ ID NO: 14
V4 variant W363N		SEQ ID NO: 15
V4 variant I356D		SEQ ID NO: 16
V4 variant I356N/G362N/W363I		SEQ ID NO: 17
V5 variant R526R/K553R		SEQ ID NO: 18
V5 variant V470F/K553R		SEQ ID NO: 19

The EIAV RT according to the invention comprises two subunits that are both encoded by a variant of the polynucleotide sequence of SEQ ID NO: 1. Variants according to the invention are described in more detail herein above. Preferably, the RT is a heterodimer. According to some embodiments, a first subunit of the two subunits comprises the complete amino acid sequence encoded by the variant and a second subunit of the two subunits comprises a proteolytic fragment of the complete amino acid sequence encoded by the variant, i.e. a proteolytic fragment of the first subunit. The first subunit has a molecular weight of 66 kDa and the second subunit has a molecular weight of 51 kDa. Expression of the open reading frame of SEQ ID NO:1 in a host cell such as *E. coli* gives rise to both the 66 kD and the 51 kD subunits that can form homodimers (i.e., p66/p66 and p51/

p51) and heterodimers (i.e., p66/p51). Some commercial preparations of RTs are mixtures of the homo- and heterodimers. While both types of dimers are active in DNA synthesis, the heterodimer presents with higher stability and processivity than the homodimers. Thus, in a preferred embodiment, the heterodimeric form of EIAV RT is purified to homogeneity.

In another aspect, the invention provides an RT obtainable by expression of the variant of the polynucleotide sequence of SEQ ID NO:1 or a vector comprising said variant in a host cell. Host cells that can be used to produce the EIAV RT according to the invention are mammalian cells, insect cells, yeast cells or bacterial cells. In a preferred embodiment, the host cells are mammalian cells such as HEK 293 or CHO cells or bacterial cells such as *E. coli*. The preferred host cells are *E. coli*.

Another aspect of the invention relates to a polynucleotide encoding an RT of the invention. Said polynucleotide is a variant of the polynucleotide sequence of SEQ ID NO:1 as described in more detail hereinabove. A further aspect of the invention relates to a vector comprising the polynucleotide according to the invention. Said polynucleotide is a variant of the polynucleotide sequence of SEQ ID NO:1 as described in more detail hereinabove. In another aspect, the invention relates to transformed host cells comprising said vector.

In yet another embodiment, the invention relates to a method of producing an RT of the invention comprising isolating the RT from the transformed host cell.

This invention thus provides novel engineered reverse transcriptase enzymes that afford beneficial improvements in thermal stability, processivity and thus length of cDNAs synthesized using an RNA template, cDNA yields and elimination of secondary enzymatic activity, i.e. RNase H activity. These improvements were incorporated into the parental (EIAV RT) molecule by directed mutagenesis of amino acid residues residing in thumb, connection, and/or RNase H domains. These mutations collectively eliminate RNase H activity and increase RT activity at elevated temperature and were identified from both rational design and random mutagenesis followed by screening. Combining these mutations allows significant improvement in performance including the capability to synthesize cDNAs of greater than 12 kb in length at the temperatures up to 65° C. and enables superior performance with even the most challenging RNA samples.

Methods

In another aspect, the present invention refers to methods for amplifying template nucleic acids comprising contacting the template nucleic acids with an RT according to the invention.

Template nucleic acids according to the present invention may be any type of nucleic acids, such as RNA, DNA, or RNA:DNA hybrids. Template nucleic acids may either be artificially produced (e.g. by molecular or enzymatic manipulations or by synthesis) or may be a naturally occurring DNA or RNA. In some preferred embodiments, the template nucleic acids are RNA sequences, such as transcription products, RNA viruses, or rRNA.

In some embodiments, the method referred to herein is RT-PCR. RT-PCR may be quantitative RT-PCR (RT-qPCR), real-time RT-PCR, digital RT-PCR (dRT-PCR) or digital droplet RT-PCR (ddRT-PCR).

In some preferred embodiments, the method of the invention comprises the steps of

- generating cDNA using a RT of the invention; and
- amplifying the generated cDNA using a DNA polymerase such as a Taq DNA polymerase.

In some embodiments, serum albumin is added during amplification, preferably recombinant human at a concentration of 1 mg/ml.

In some embodiments, the method of the invention further comprises detecting and/or quantifying the amplified nucleic acids. Quantification/detection of amplified nucleic acids may be performed, e.g., using non-sequence-specific fluorescent dyes (e.g., SYBR® Green, EvaGreen®) that intercalate into double-stranded DNA molecules in a sequence non-specific manner, or sequence-specific DNA probes (e.g., oligonucleotides labelled with fluorescent reporters) that permit detection only after hybridization with the DNA targets, synthesis-dependent hydrolysis or after incorporation into PCR products.

Kits

Reagents necessary to perform the method of the invention may be comprised in kits.

In some embodiments, the invention relates to kits for amplifying template nucleic acids, wherein the kit comprises an RT of the invention and a buffer. Optionally, the kit additionally comprises Taq DNA polymerase and/or serum albumin. Buffers comprised in the kit may be conventional buffers containing magnesium.

EXAMPLES

The invention is illustrated in the following examples.

Example 1: Expression and Purification of EIAV RT

The open reading frame of SEQ ID NO:1 or a variant thereof was expressed in *E. coli* which gives rise to a 66 kDa polypeptide and a proteolytically processed fragment thereof, i.e. a 51 kDa polypeptide. The 66 kDa polypeptide has an amino acid sequence that is at least 90% identical to the amino acid sequence of SEQ ID NO:2. The 66 kDa and 51 kDa polypeptides form homodimers (i.e., p66/p66 and p51/p51) and heterodimers (i.e., p66/p51). In one preparation, referred to as the “homogeneous” preparation, the heterodimeric form (p66/p51) was separated from either of the homodimeric forms (p66/p66 and p51/p51) through multiple rounds of chromatography steps including heparin, hydrophobic interaction, and ion exchange chromatography; in a second preparation, referred to as the “heterogeneous” preparation, the homodimers and heterodimers were present in approximately equal amounts. While both homodimers and heterodimers are active in DNA synthesis, homogeneous preparations of the heterodimer provide higher stability and processivity than heterogenous mixes of homodimers and heterodimers, as well as improved performance (FIGS. 1 and 2). Unless indicated otherwise, all of RT variants in the following examples were tested as homogeneous preparations of heterodimers.

Example 2: Generation of EIAV RT Mutants

RNase H^{neg} Mutations

Based on homologous mutations that eliminated RNase H activity of commonly studied RTs, i.e. MMLV and HIV-1 RT (Mizuno 2010, Gerard 2002), mutation of the EIAV RT RNase H domain was expected to enhance thermal stability.

Thus, RNase H^{neg} variants (EIAV RT V1 and V2; see Table 3 below) of EIAV RT were generated by site directed mutagenesis. When samples of up to 250 ng were tested, neither EIAV RT V1 nor EIAV RT V2 showed detectable RNase H activity (data not shown).

Thumb Domain Mutations

The minor groove binding track (MGBT) in the thumb domain is a highly conserved structural element among retroviral RTs (Beard 1998, Bebenek 1997). Biochemical and molecular modeling studies of HIV-1 RT have revealed that this element is critical for maintaining reading frames, fidelity and processivity by increasing template-primer binding affinity (Beard 1994, Bebenek 1995). Based on sequence alignment of the thumb domain of other lentiviral RTs, five point mutations (M263V/N265K/T267N/M269A/S271Q) within MGBT (EIAV RT V3) were introduced into EIAV RT V1 by site directed mutagenesis.

Random Mutations and Screening on Connection and RNase H Domain.

Key mutations that improve thermostability of RTs have been identified in connection and RNase H domains (Mata-moros 2013). To further improve thermostability and processivity, random mutation libraries of connection and RNase H domains were constructed and screened. Based on crystal structures of HIV-1 RT complexed with template/primer and on modeling studies (Huang 1998), a region containing 10 amino acids (355-364) in the connection domain (CD) was chosen for randomization. A CD library was generated by bridging dsDNA with a mixture of 10 ssDNA oligos containing one random amino acid at each position. A random library of RNase H domain variants was generated by error prone PCR. A screening assay of the libraries was performed by measuring RT activity of heat treated (50° C. for 10 min) crude cell lysate at 60° C. The results identified four variants from the CD library and three variants from the RNase H domain library that showed higher RT activity than the parent EIAV RT V3 (data not shown). The best performing CD variant was EIAV RT V4 and the best performing RNase H domain variant was EIAV RT V5. The mutations of EIAV RT V4 and EIAV RT V5 were combined to generate the variant EIAV RT V6.

TABLE 3

EIAV best mode constructs			
Variant	SEQ ID	Mutation	Description of the Effect
EIAV RT WT	SEQ ID NO 2	none	Heterodimeric wild type enzyme
EIAV RT V1	SEQ ID NO 4	D443G	Heterodimeric enzyme mutagenized to eliminate RNase H activity
EIAV RT V2	SEQ ID NO 6	D443G/E476Q	Heterodimeric enzyme mutagenized to eliminate RNase H activity
EIAV RT V3	SEQ ID NO 8	D443G/M263V/N265K/T267N/M269A/S271Q	Heterodimeric enzyme mutagenized to eliminate RNase H activity and modify the MGBT in thumb domain to increase affinity for template
EIAV RT V4	SEQ ID NO 10	D443G/M263V/N265K/T267N/M269A/S271Q/I356G/N360A/G362T/W363K	MGBT mutant with additional CD mutations

TABLE 3-continued

EIAV best mode constructs			
Variant	SEQ ID	Mutation	Description of the Effect
EIAV RT V5	SEQ ID NO 12	D443G/M263V/N265K/T267N/M269A/S271Q/Q491R/R526H	MGBT mutant with additional RNase H domain mutations
EIAV RT V6	SEQ ID NO 14	D443G/M263V/N265K/T267N/M269A/S271Q/I356G/N360A/G362T/W363K/Q491R/R526H	MGBT mutant with additional CD and RNase H domain mutations

Example 3: Assessment of Thermostability and Thermal Activity of RNase H^{neg} Variants

Thermostability and thermal activity of the RNase H variants EIAV RT V1 and V2 were evaluated by thermal inactivation assay (FIG. 1) and first strand cDNA synthesis (FIG. 2). In the thermal inactivation assay, RTs were preincubated with a substrate (Oligo (dT)20 primed Poly (A) template) at the 42-60° C. for 10 min. Following preincubation, RT reaction was initiated by adding MgCl₂ and the activities were measured by monitoring the relative rates of dT incorporation at 42° C. All the RNase H^{neg} variants (EIAV RT V1 and V2) remained fully active after preincubation at 50° C. whereas heterogeneous EIAV RT WT and homogeneous EIAV RT WT showed 50 and 30% reduced activities, respectively (FIG. 1). Notably, double H^{neg} mutant, EIAV RT V2 retained 80% of its activity even at 55° C. preincubation.

The thermostability of these variants was more stringently evaluated by measuring cDNA synthesis at elevated temperature. First strand cDNA synthesis reactions were performed at 50° C. using a mixture of RNAs ranging in length from 0.5 to 9 kb as template (poly(A)-tailed RNA ladder). Single strand cDNA was resolved by alkaline electrophoresis and visualized by staining with SYBR Gold. As shown in FIG. 2(a), all RNase H^{neg} variants efficiently synthesized full-length cDNAs up to 9 kb, while EIAV RT WT exhibited lower cDNA yield than RNase H^{neg} variants (Lane 1: heterogeneous EIAV RT, Lane 2: homogeneous EIAV RT WT, Lane 3: EIAV RT V1, Lane 4: EIAV RT V2). In contrast, full-length products of any size from heterogeneous EIAV RT WT are near the limit of visual detection, while homogeneous EIAV RT WT yield was slightly lower than the RNase H^{neg} variants.

In a 2-step RT-PCR assay, initial cDNA synthesis reactions were done at 50° C. (FIG. 2(b), upper panel) or 60° C. (FIG. 2(b), lower panel) for 30 min using 10 ng of human total RNA as template. A 2 kb fragment of APC gene was then amplified. All RNase H^{neg} variants generated amplifiable cDNA at both 50° C. and 60° C. whereas homogeneous and heterogeneous EIAV RT both failed to generate cDNA at either 50° C. or 60° C. (FIG. 2(b)). Taken together, these results collectively support the interpretation that abolishing RNase H activity of RT increases its thermostability.

Example 4: Assessment of Thermostability and Thermal Activity of an RNase H^{neg} EIAV RT Variant with Additional Mutations within the MGBT of the Thumb Domain (EIAV RT V3)

The effects of the additional mutations on thermostability and thermal activity were evaluated using a FAM probe-based one-step RT-qPCR. First, a region of the ACTB gene

was reverse transcribed from human total RNA (2 pg) using EIAV RT V1, EIAV RT V3 or EnzScript RT as indicated, and then amplified and quantified. The results show that both EIAV RT V1 and EIAV RT V3 retain most of their activity close to 70° C. while showing optimal activities at temperature around 55° C. However, reduced activity of EIAV RT V1 becomes noticeable at temperatures higher than 65° C. In contrast, substantial increases of Ct values, denoting reduced cDNA yields, were observed with a typical, widely used comparator EnzScript (a Moloney murine leukemia virus (MMLV) RT with reduced RNase H activity) as temperature increases (FIG. 3).

Additionally, a 2-step RT-qPCR assay was performed using EIAV RT V1, EIAV RT V3 and SuperScript IV. Using human total RNA (50 ng) and a gene specific primer (MAP4), cDNA was reverse transcribed at the indicated temperatures and cDNAs of different lengths were quantified by qPCR targeting the MAP4 gene. The results demonstrate that EIAV RT V3 generates higher cDNA yield, as shown by lower Ct value, than EIAV RT V1. The difference in yield between these enzymes was dramatically increased at higher temperature or for longer cDNA synthesis indicating improvement of thermostability and processivity of EIAV RT V3 compared with EIAV RT V1 (FIG. 4).

The speed of synthesis and length of product for cDNAs generated by the EIAV RT V3 were compared to SuperScript IV, a widely used commercial RT that is considered state of the art. The reaction was performed using 50 ng of human total RNA as input at 55° C. and Oligo (dT)20 as a primer. The reaction was quenched by heat deactivating RTs at 85° C. for 5 min at indicated times followed by amplification of a 9 kb APC gene. Both enzymes generated amplifiable 9 kb products within 10 min. However, yields were substantially higher with EIAV RT V3 (FIG. 5).

Example 5: Assessment of Thermostability and Thermal Activity of RNase H^{neg}/MGBT^{mut} EIAV RT Variants with Additional Mutations within the Connection Domain and/or RNase H Domain

The effects of additional mutations within the CD and/or RNase H domain on thermostability and thermal activity were tested in 2 step RT-qPCR (FIG. 6) and a long-range endpoint PCR (FIG. 7).

For the 2 step RT-qPCR, cDNA synthesis using human total RNA (50 ng) and a gene specific primer (MAP4) was performed at the indicated temperatures. Then, cDNA of different lengths were quantified by amplifying MAP4 gene using primer sets designed for the specific regions of the gene. For the long range 2-step RT-PCR (12.3 kb) total rat brain RNA (50 ng) and a gene specific primer for rat dynein were used for the cDNA synthesis at different temperatures (FIG. 7; Lane 1: EIAV RT V3, Lane 2: EIAV RT V4, Lane 3: EIAV RT V5, Lane 4: EIAV RT V6).

Both 2 step RT-qPCR (FIG. 6) and long-range endpoint PCR (FIG. 7) assays at different temperature confirm that all of these variants (EIAV RT V4, EIAV RT V5 and EIAV RT V6) showed increased thermostability and processivity compared with the original EIAV RT V3. One of the constructs, EIAV RT V4, was capable of synthesizing 12.3 kb cDNA at temperatures as high as 65° C. (FIG. 7), and this variant is considered best mode for most applications.

SEQUENCE LISTING

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Lys Leu Glu Gly Ala Lys Glu Ile Val Gln Arg Leu Leu Ser Glu Gly
 35           40           45

Lys Ile Ser Glu Ala Ser Asp Asn Asn Pro Tyr Asn Ser Pro Ile Phe
 50           55           60

Val Ile Lys Lys Arg Ser Gly Lys Trp Arg Leu Leu Gln Asp Leu Arg
 65           70           75           80

Glu Leu Asn Lys Thr Val Gln Val Gly Thr Glu Ile Ser Arg Gly Leu
 85           90           95

Pro His Pro Gly Gly Leu Ile Lys Cys Lys His Met Thr Val Leu Asp
100           105           110

Ile Gly Asp Ala Tyr Phe Thr Ile Pro Leu Asp Pro Glu Phe Arg Pro
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Tyr Thr Ala Phe Thr Ile Pro Ser Ile Asn His Gln Glu Pro Asp Lys
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Arg Tyr Val Trp Asn Cys Leu Pro Gln Gly Phe Val Leu Ser Pro Tyr
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Tyr Pro Glu Val Gln Leu Tyr Gln Tyr Met Asp Asp Leu Phe Val Gly
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Ser Asn Gly Ser Lys Lys Gln His Lys Glu Leu Ile Ile Glu Leu Arg
195           200           205

Ala Ile Leu Leu Glu Lys Gly Phe Glu Thr Pro Asp Asp Lys Leu Gln
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Glu Val Pro Pro Tyr Ser Trp Leu Gly Tyr Gln Leu Cys Pro Glu Asn
225           230           235           240

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245           250           255

Asn Asp Val Gln Lys Leu Met Gly Asn Ile Thr Trp Met Ser Ser Gly
260           265           270

Val Pro Gly Leu Thr Val Lys His Ile Ala Ala Thr Thr Lys Gly Cys
275           280           285

Leu Glu Leu Asn Gln Lys Val Ile Trp Thr Glu Glu Ala Gln Lys Glu
290           295           300

Leu Glu Glu Asn Asn Glu Lys Ile Lys Asn Ala Gln Gly Leu Gln Tyr
305           310           315           320

Tyr Asn Pro Glu Glu Glu Met Leu Cys Glu Val Glu Ile Thr Lys Asn
325           330           335

Tyr Glu Ala Thr Tyr Val Ile Lys Gln Ser Gln Gly Ile Leu Trp Ala
340           345           350

Gly Lys Lys Ile Met Lys Ala Asn Lys Gly Trp Ser Thr Val Lys Asn
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Asp Ser Tyr Tyr Cys Trp Lys Asn Ile Thr Glu Gly Leu Gly Leu Glu		
500	505	510
Gly Pro Gln Ser Pro Trp Trp Pro Ile Ile Gln Asn Ile Arg Glu Lys		
515	520	525
Glu Ile Val Tyr Phe Ala Trp Val Pro Gly His Lys Gly Ile Cys Gly		
530	535	540
Asn Gln Leu Ala Asp Glu Ala Ala Lys Ile Lys Glu Glu Ile		
545	550	555

<210> SEQ ID NO 5
 <211> LENGTH: 1674
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <221> NAME/KEY: source
 <222> LOCATION: 1..1674
 <223> OTHER INFORMATION: /mol_type="unassigned DNA"
 /note="Synthetic gene"
 /organism="Artificial Sequence"

<400> SEQUENCE: 5

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ggtccgaaaa ttccgcagtg gcctctgacc aaagaaaaac tggaaggtgc aaaagaaatt	120
gttcagcgtc tgctgagcga aggtaaaatt agcgaagcaa gcgataataa cccgtataac	180
agcccgattt tcgtgatcaa aaaacgtagc ggtaaatggc gtctgctgca ggatctgcgt	240
gaactgaata aaaccgttca ggttggcacc gaaattagcc gtggtctgcc gcatccgggt	300

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ggcttgatta aatgtaaaca catgaccgtt ctggatattg gcgacgcata ttttaccatt 360
ccgctggatc cggaatttcg tccgtatacc gcatttaca ttccgagcat taatcatcaa 420
gaaccggata aacgttatgt gtggaattgc ctgccgcagg gttttgttct gagcccgat 480
atctatcaga aaacgctgca agaaattctg cagccatttc gtgaacgtta tccggaagtt 540
cagctgtatc agtatatgga tgacctgttt gtgggtagca atggtagcaa aaaacagcac 600
aaagaactga tcattgaact gcgtgcaatt ctgctggaaa aaggttttga aacaccggat 660
gataaactgc aagaagttcc gccttatagc tggctgggtt atcagctgtg tccggaaaat 720
tggaagttc agaaaatgca gctggatatg gttaaaaatc cgacctgaa tgatgtgcag 780
aaactgatgg gcaatattac ctggatgagc agcggtggtc cgggtctgac cgttaaacat 840
attgcagcaa ccaccaaagg ttgtctggaa ctgaatcaga aagttatctg gaccgaagaa 900
gcacaaaaag aactggaaga aaacaacgag aaaatcaaaa atgcacaggg cctgcagtat 960
tataacccgg aagaagaat gctttgcgaa gtggaatca ccaaaaacta tgaagccacc 1020
tatgtgatta aacagagcca gggatttctg tgggcaggca aaaaaatcat gaaagccaat 1080
aaaggttgga gcaccgtgaa aaatctgatg ctgctgctgc aacatgtggc aaccgaaagc 1140
attacccgtg ttggtaaatg tccgaccttt aaagtccgt ttaccaaaga acaggtgatg 1200
tggaagtgc agaaaggtg gtattatagt tggctgccgg aaattgttta taccatcag 1260
gtgttcctat atgactggcg tatgaaactg gttgaagaac cgaccagcgg tattaccatt 1320
tataccggtg gtggtaaaca gaattgtgaa ggtattgcag cctatgttac cagcaatggt 1380
cgtaccaaac agaaacgtct gggtcgggtt acacatcagg tggcacaacg tatggcaatt 1440
cagatggcac tggaagatac ccgtgataaa caggtaata ttgtgaccga tagctattat 1500
tggtggaaaa acattaccga aggtctgggt ctggaaggtc cgcagagccc gtggtggccg 1560
attattcaga atattcgcga aaaagaaatc gtgtactttg catgggttcc aggtcataaa 1620
ggtatttgtg gtaatcagct ggcagatgaa gcagccaaaa tcaaagaaga gatc 1674

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<210> SEQ ID NO 6

<211> LENGTH: 558

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Expression from a synthetic gene

<400> SEQUENCE: 6

```

Met Ala Gln Leu Ser Lys Glu Ile Lys Phe Arg Lys Ile Glu Leu Lys
1           5           10           15

```

```

Glu Gly Thr Met Gly Pro Lys Ile Pro Gln Trp Pro Leu Thr Lys Glu
20           25           30

```

```

Lys Leu Glu Gly Ala Lys Glu Ile Val Gln Arg Leu Leu Ser Glu Gly
35           40           45

```

```

Lys Ile Ser Glu Ala Ser Asp Asn Asn Pro Tyr Asn Ser Pro Ile Phe
50           55           60

```

```

Val Ile Lys Lys Arg Ser Gly Lys Trp Arg Leu Leu Gln Asp Leu Arg
65           70           75           80

```

```

Glu Leu Asn Lys Thr Val Gln Val Gly Thr Glu Ile Ser Arg Gly Leu
85           90           95

```

```

Pro His Pro Gly Gly Leu Ile Lys Cys Lys His Met Thr Val Leu Asp
100          105          110

```

```

Ile Gly Asp Ala Tyr Phe Thr Ile Pro Leu Asp Pro Glu Phe Arg Pro

```

-continued

115	120	125
Tyr Thr Ala Phe Thr Ile Pro Ser Ile Asn His Gln Glu Pro Asp Lys 130 135 140		
Arg Tyr Val Trp Asn Cys Leu Pro Gln Gly Phe Val Leu Ser Pro Tyr 145 150 155 160		
Ile Tyr Gln Lys Thr Leu Gln Glu Ile Leu Gln Pro Phe Arg Glu Arg 165 170 175		
Tyr Pro Glu Val Gln Leu Tyr Gln Tyr Met Asp Asp Leu Phe Val Gly 180 185 190		
Ser Asn Gly Ser Lys Lys Gln His Lys Glu Leu Ile Ile Glu Leu Arg 195 200 205		
Ala Ile Leu Leu Glu Lys Gly Phe Glu Thr Pro Asp Asp Lys Leu Gln 210 215 220		
Glu Val Pro Pro Tyr Ser Trp Leu Gly Tyr Gln Leu Cys Pro Glu Asn 225 230 235 240		
Trp Lys Val Gln Lys Met Gln Leu Asp Met Val Lys Asn Pro Thr Leu 245 250 255		
Asn Asp Val Gln Lys Leu Met Gly Asn Ile Thr Trp Met Ser Ser Gly 260 265 270		
Val Pro Gly Leu Thr Val Lys His Ile Ala Ala Thr Thr Lys Gly Cys 275 280 285		
Leu Glu Leu Asn Gln Lys Val Ile Trp Thr Glu Glu Ala Gln Lys Glu 290 295 300		
Leu Glu Glu Asn Asn Glu Lys Ile Lys Asn Ala Gln Gly Leu Gln Tyr 305 310 315 320		
Tyr Asn Pro Glu Glu Glu Met Leu Cys Glu Val Glu Ile Thr Lys Asn 325 330 335		
Tyr Glu Ala Thr Tyr Val Ile Lys Gln Ser Gln Gly Ile Leu Trp Ala 340 345 350		
Gly Lys Lys Ile Met Lys Ala Asn Lys Gly Trp Ser Thr Val Lys Asn 355 360 365		
Leu Met Leu Leu Leu Gln His Val Ala Thr Glu Ser Ile Thr Arg Val 370 375 380		
Gly Lys Cys Pro Thr Phe Lys Val Pro Phe Thr Lys Glu Gln Val Met 385 390 395 400		
Trp Glu Met Gln Lys Gly Trp Tyr Tyr Ser Trp Leu Pro Glu Ile Val 405 410 415		
Tyr Thr His Gln Val Val His Asp Asp Trp Arg Met Lys Leu Val Glu 420 425 430		
Glu Pro Thr Ser Gly Ile Thr Ile Tyr Thr Gly Gly Gly Lys Gln Asn 435 440 445		
Gly Glu Gly Ile Ala Ala Tyr Val Thr Ser Asn Gly Arg Thr Lys Gln 450 455 460		
Lys Arg Leu Gly Pro Val Thr His Gln Val Ala Gln Arg Met Ala Ile 465 470 475 480		
Gln Met Ala Leu Glu Asp Thr Arg Asp Lys Gln Val Asn Ile Val Thr 485 490 495		
Asp Ser Tyr Tyr Cys Trp Lys Asn Ile Thr Glu Gly Leu Gly Leu Glu 500 505 510		
Gly Pro Gln Ser Pro Trp Trp Pro Ile Ile Gln Asn Ile Arg Glu Lys 515 520 525		
Glu Ile Val Tyr Phe Ala Trp Val Pro Gly His Lys Gly Ile Cys Gly 530 535 540		

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Asn Gln Leu Ala Asp Glu Ala Ala Lys Ile Lys Glu Glu Ile
545 550 555

<210> SEQ ID NO 7
<211> LENGTH: 1674
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<222> LOCATION: 1..1674
<223> OTHER INFORMATION: /mol_type="unassigned DNA"
/note="Synthetic gene"
/organism="Artificial Sequence"

<400> SEQUENCE: 7

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gttcagcgtc tgctgagcga aggtaaaatt agcgaagcaa gcgataataa cccgtataac    180
agcccgattt tcgtgatcaa aaaacgtagc ggtaaatggc gtctgctgca ggatctgcgt    240
gaactgaata aaacggttca ggttggcacc gaaattagcc gtggtctgcc gcacccgggt    300
ggtctgatta aatgtaaaca catgaccgtt ctggatattg gcgacgcata ttttaccatt    360
ccgtggatc cggaatttcg tccgtatacc gcatttaca ttccgagcat taatcatcaa    420
gaaccggata aacgttatgt gtggaattgc ctgccgcagg gttttgttct gagcccgat    480
atctatcaga aaacgctgca agaaattctg cagccatttc gtgaacgtta tccggaagtt    540
cagctgtatc agtatatgga tgacctgttt gtgggtagca atggtagcaa aaaacagcac    600
aaagaactga tcattgaact gcgtgcaatt ctgctggaaa aaggttttga aacaccggat    660
gataaactgc aagaagttcc gccttatagc tggctgggtt atcagctgtg tccggaatat    720
tggaagttc agaaaatgca gctggatatg gttaaaaatc cgaccctgaa tgatgtgcag    780
aaactgggtg gcaaaattaa ttgggctagc cagggtgttc cgggtctgac cgtaaacat    840
attgcagcaa ccaccaaagg ttgtctggaa ctgaatcaga aagttatctg gaccgaagaa    900
gcacaaaaag aactggaaga aaacaacgag aaaatcaaaa atgcacaggg cctgcagtat    960
tataccccgg aagaagaat gctttgcgaa gtggaaatca ccaaaaacta tgaagccacc   1020
tatgtgatta aacagagcca gggatttctg tgggcaggca aaaaaatcat gaaagccaat   1080
aaaggttgga gcaccgtgaa aaatctgatg ctgctgctgc aacatgtggc aaccgaaagc   1140
attacccgtg ttggtaaatg tccgacctt aaagtccgt ttaccaaaga acaggtgatg   1200
tggaagtgc agaaaggttg gtattatagt tggctgccg aaattgttta taccatcag   1260
gttgttcatg atgactggcg tatgaaactg gttgaagaac cgaccagcgg tattaccatt   1320
tataccggtg gtggtaaaca gaatgtgaa ggtattgcag cctatgttac cagcaatggt   1380
cgtaccaaac agaaacgtct gggtcgggtt acacatcagg tggcagaacg tatggcaatt   1440
cagatggcac tggaagatac ccgtgataaa caggtaataa ttgtgaccga tagctattat   1500
tgttgaaaaa acattaccga aggtctgggt ctggaaggtc cgacagagcc gtggtggccg   1560
attattcaga atattcgcga aaaagaaatc gtgtactttg catgggttcc aggtcataaa   1620
ggtatttgtg gtaatcagct ggcagatgaa gcagccaaaa tcaaagaaga gatc      1674
```

<210> SEQ ID NO 8
<211> LENGTH: 558
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence

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<220> FEATURE:

<223> OTHER INFORMATION: Expression from a synthetic gene

<400> SEQUENCE: 8

```

Met Ala Gln Leu Ser Lys Glu Ile Lys Phe Arg Lys Ile Glu Leu Lys
1           5           10           15

Glu Gly Thr Met Gly Pro Lys Ile Pro Gln Trp Pro Leu Thr Lys Glu
20           25           30

Lys Leu Glu Gly Ala Lys Glu Ile Val Gln Arg Leu Leu Ser Glu Gly
35           40           45

Lys Ile Ser Glu Ala Ser Asp Asn Asn Pro Tyr Asn Ser Pro Ile Phe
50           55           60

Val Ile Lys Lys Arg Ser Gly Lys Trp Arg Leu Leu Gln Asp Leu Arg
65           70           75           80

Glu Leu Asn Lys Thr Val Gln Val Gly Thr Glu Ile Ser Arg Gly Leu
85           90           95

Pro His Pro Gly Gly Leu Ile Lys Cys Lys His Met Thr Val Leu Asp
100          105          110

Ile Gly Asp Ala Tyr Phe Thr Ile Pro Leu Asp Pro Glu Phe Arg Pro
115          120          125

Tyr Thr Ala Phe Thr Ile Pro Ser Ile Asn His Gln Glu Pro Asp Lys
130          135          140

Arg Tyr Val Trp Asn Cys Leu Pro Gln Gly Phe Val Leu Ser Pro Tyr
145          150          155          160

Ile Tyr Gln Lys Thr Leu Gln Glu Ile Leu Gln Pro Phe Arg Glu Arg
165          170          175

Tyr Pro Glu Val Gln Leu Tyr Gln Tyr Met Asp Asp Leu Phe Val Gly
180          185          190

Ser Asn Gly Ser Lys Lys Gln His Lys Glu Leu Ile Ile Glu Leu Arg
195          200          205

Ala Ile Leu Leu Glu Lys Gly Phe Glu Thr Pro Asp Asp Lys Leu Gln
210          215          220

Glu Val Pro Pro Tyr Ser Trp Leu Gly Tyr Gln Leu Cys Pro Glu Asn
225          230          235          240

Trp Lys Val Gln Lys Met Gln Leu Asp Met Val Lys Asn Pro Thr Leu
245          250          255

Asn Asp Val Gln Lys Leu Val Gly Lys Ile Asn Trp Ala Ser Gln Gly
260          265          270

Val Pro Gly Leu Thr Val Lys His Ile Ala Ala Thr Thr Lys Gly Cys
275          280          285

Leu Glu Leu Asn Gln Lys Val Ile Trp Thr Glu Glu Ala Gln Lys Glu
290          295          300

Leu Glu Glu Asn Asn Glu Lys Ile Lys Asn Ala Gln Gly Leu Gln Tyr
305          310          315          320

Tyr Asn Pro Glu Glu Glu Met Leu Cys Glu Val Glu Ile Thr Lys Asn
325          330          335

Tyr Glu Ala Thr Tyr Val Ile Lys Gln Ser Gln Gly Ile Leu Trp Ala
340          345          350

Gly Lys Lys Ile Met Lys Ala Asn Lys Gly Trp Ser Thr Val Lys Asn
355          360          365

Leu Met Leu Leu Leu Gln His Val Ala Thr Glu Ser Ile Thr Arg Val
370          375          380

Gly Lys Cys Pro Thr Phe Lys Val Pro Phe Thr Lys Glu Gln Val Met
385          390          395          400

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-continued

Asn Asp Val Gln Lys Leu Val Gly Lys Ile Asn Trp Ala Ser Gln Gly
 260 265 270
 Val Pro Gly Leu Thr Val Lys His Ile Ala Ala Thr Thr Lys Gly Cys
 275 280 285
 Leu Glu Leu Asn Gln Lys Val Ile Trp Thr Glu Glu Ala Gln Lys Glu
 290 295 300
 Leu Glu Glu Asn Asn Glu Lys Ile Lys Asn Ala Gln Gly Leu Gln Tyr
 305 310 315 320
 Tyr Asn Pro Glu Glu Glu Met Leu Cys Glu Val Glu Ile Thr Lys Asn
 325 330 335
 Tyr Glu Ala Thr Tyr Val Ile Lys Gln Ser Gln Gly Ile Leu Trp Ala
 340 345 350
 Gly Lys Lys Gly Met Lys Ala Ala Lys Thr Lys Ser Thr Val Lys Asn
 355 360 365
 Leu Met Leu Leu Leu Gln His Val Ala Thr Glu Ser Ile Thr Arg Val
 370 375 380
 Gly Lys Cys Pro Thr Phe Lys Val Pro Phe Thr Lys Glu Gln Val Met
 385 390 395 400
 Trp Glu Met Gln Lys Gly Trp Tyr Tyr Ser Trp Leu Pro Glu Ile Val
 405 410 415
 Tyr Thr His Gln Val Val His Asp Asp Trp Arg Met Lys Leu Val Glu
 420 425 430
 Glu Pro Thr Ser Gly Ile Thr Ile Tyr Thr Gly Gly Gly Lys Gln Asn
 435 440 445
 Gly Glu Gly Ile Ala Ala Tyr Val Thr Ser Asn Gly Arg Thr Lys Gln
 450 455 460
 Lys Arg Leu Gly Pro Val Thr His Gln Val Ala Glu Arg Met Ala Ile
 465 470 475 480
 Gln Met Ala Leu Glu Asp Thr Arg Asp Lys Gln Val Asn Ile Val Thr
 485 490 495
 Asp Ser Tyr Tyr Cys Trp Lys Asn Ile Thr Glu Gly Leu Gly Leu Glu
 500 505 510
 Gly Pro Gln Ser Pro Trp Trp Pro Ile Ile Gln Asn Ile Arg Glu Lys
 515 520 525
 Glu Ile Val Tyr Phe Ala Trp Val Pro Gly His Lys Gly Ile Cys Gly
 530 535 540
 Asn Gln Leu Ala Asp Glu Ala Ala Lys Ile Lys Glu Glu Ile
 545 550 555

<210> SEQ ID NO 11
 <211> LENGTH: 1674
 <212> TYPE: DNA
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <221> NAME/KEY: source
 <222> LOCATION: 1..1674
 <223> OTHER INFORMATION: /mol_type="unassigned DNA"
 /note="Synthetic gene"
 /organism="Artificial Sequence"

<400> SEQUENCE: 11

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ggtccgaaaa ttccgcagtg gcctctgacc aaagaaaaac tggaagggtgc aaaagaaatt	120
gttcagcgtc tgctgagcga aggtaaaatt agcgaagcaa gcgataataa cccgtataac	180
agcccgatatt tcgtgatcaa aaaacgtagc ggtaaatggc gtctgctgca ggatctgcgt	240

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gaactgaata aaaccgttca ggttggcacc gaaattagcc gtggtctgcc gcacccgggt    300
ggtctgatta aatgtaaaca catgaccgtt ctggatattg gcgacgcata ttttaccatt    360
ccgctggatc cggaatttcg tccgtatacc gcatttaca tcccgagcat taatcatcaa    420
gaaccggata aacgttatgt gtggaattgc ctgccgcagg gttttgttct gagcccgat    480
atctatcaga aaacgctgca agaaattctg cagccatttc gtgaacgtta tccggaagtt    540
cagctgtatc agtatatgga tgacctgttt gtgggtagca atggtagcaa aaaacagcac    600
aaagaactga tcattgaact gcgtgcaatt ctgctggaaa aaggttttga aacaccggat    660
gataaactgc aagaagtccc gccttatagc tggtctgggt atcagctgtg tccgaaaaat    720
tggaagtcc agaaaatgca gctggatatg gttaaaaatc cgaccctgaa tgatgtgcag    780
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gcacaaaaag aactggaaga aaacaacgag aaatcaaaa atgcacaggg cctgcagtat    960
tataaccggg aagaagaaat gctttgcgaa gtggaatca ccaaaaacta tgaagccacc   1020
tatgtgatta aacagagcca gggatattctg tgggcaggca aaaaaatcat gaaagccaat   1080
aaaggttgga gcaccgtgaa aaatctgatg ctgctgtctg aacatgtggc aaccgaaagc   1140
attaccctg ttggtaaatg tccgaccttt aaagttccgt ttaccaaaga acaggtgatg   1200
tggaagtgc agaaaggttg gtattatagt tggtgccgg aaattgttta taccatcag   1260
gtgttcctat atgactggcg tatgaaactg gttgaagaac cgaccagcgg tattaccatt   1320
tataccgggt gtggtaaaca gaatggtgaa ggtattgcag cctatgttac cagcaatggt   1380
cgtaccaaac agaaacgtct gggtcgggtt acacatcagg tggcagaacg tatggcaatt   1440
cagatggcac tggaagatac ccgtgataaa cgggttaata ttgtgaccga tagctattat   1500
tggtggaaaa acattaccga aggtctgggt ctggaaggtc cgcagagccc gtggtggccc   1560
attattcaga atattcacga aaaagaaatc gtgtactttg catgggttcc aggtcataaa   1620
ggtatttgtg gtaatcagct ggcagatgaa gcagccaaaa tcaaagaaga gatc       1674

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<210> SEQ ID NO 12

<211> LENGTH: 558

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Expression from a synthetic gene

<400> SEQUENCE: 12

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Met Ala Gln Leu Ser Lys Glu Ile Lys Phe Arg Lys Ile Glu Leu Lys
1           5           10           15

```

```

Glu Gly Thr Met Gly Pro Lys Ile Pro Gln Trp Pro Leu Thr Lys Glu
20           25           30

```

```

Lys Leu Glu Gly Ala Lys Glu Ile Val Gln Arg Leu Leu Ser Glu Gly
35           40           45

```

```

Lys Ile Ser Glu Ala Ser Asp Asn Asn Pro Tyr Asn Ser Pro Ile Phe
50           55           60

```

```

Val Ile Lys Lys Arg Ser Gly Lys Trp Arg Leu Leu Gln Asp Leu Arg
65           70           75           80

```

```

Glu Leu Asn Lys Thr Val Gln Val Gly Thr Glu Ile Ser Arg Gly Leu
85           90           95

```

```

Pro His Pro Gly Gly Leu Ile Lys Cys Lys His Met Thr Val Leu Asp
100          105          110

```

-continued

Ile Gly Asp Ala Tyr Phe Thr	Ile Pro Leu Asp Pro Glu Phe Arg Pro
115	120 125
Tyr Thr Ala Phe Thr Ile Pro Ser Ile Asn His Gln Glu Pro Asp Lys	
130	135 140
Arg Tyr Val Trp Asn Cys Leu Pro Gln Gly Phe Val Leu Ser Pro Tyr	
145	150 155 160
Ile Tyr Gln Lys Thr Leu Gln Glu Ile Leu Gln Pro Phe Arg Glu Arg	
165	170 175
Tyr Pro Glu Val Gln Leu Tyr Gln Tyr Met Asp Asp Leu Phe Val Gly	
180	185 190
Ser Asn Gly Ser Lys Lys Gln His Lys Glu Leu Ile Ile Glu Leu Arg	
195	200 205
Ala Ile Leu Leu Glu Lys Gly Phe Glu Thr Pro Asp Asp Lys Leu Gln	
210	215 220
Glu Val Pro Pro Tyr Ser Trp Leu Gly Tyr Gln Leu Cys Pro Glu Asn	
225	230 235 240
Trp Lys Val Gln Lys Met Gln Leu Asp Met Val Lys Asn Pro Thr Leu	
245	250 255
Asn Asp Val Gln Lys Leu Val Gly Lys Ile Asn Trp Ala Ser Gln Gly	
260	265 270
Val Pro Gly Leu Thr Val Lys His Ile Ala Ala Thr Thr Lys Gly Cys	
275	280 285
Leu Glu Leu Asn Gln Lys Val Ile Trp Thr Glu Glu Ala Gln Lys Glu	
290	295 300
Leu Glu Glu Asn Asn Glu Lys Ile Lys Asn Ala Gln Gly Leu Gln Tyr	
305	310 315 320
Tyr Asn Pro Glu Glu Glu Met Leu Cys Glu Val Glu Ile Thr Lys Asn	
325	330 335
Tyr Glu Ala Thr Tyr Val Ile Lys Gln Ser Gln Gly Ile Leu Trp Ala	
340	345 350
Gly Lys Lys Ile Met Lys Ala Asn Lys Gly Trp Ser Thr Val Lys Asn	
355	360 365
Leu Met Leu Leu Leu Gln His Val Ala Thr Glu Ser Ile Thr Arg Val	
370	375 380
Gly Lys Cys Pro Thr Phe Lys Val Pro Phe Thr Lys Glu Gln Val Met	
385	390 395 400
Trp Glu Met Gln Lys Gly Trp Tyr Tyr Ser Trp Leu Pro Glu Ile Val	
405	410 415
Tyr Thr His Gln Val Val His Asp Asp Trp Arg Met Lys Leu Val Glu	
420	425 430
Glu Pro Thr Ser Gly Ile Thr Ile Tyr Thr Gly Gly Gly Lys Gln Asn	
435	440 445
Gly Glu Gly Ile Ala Ala Tyr Val Thr Ser Asn Gly Arg Thr Lys Gln	
450	455 460
Lys Arg Leu Gly Pro Val Thr His Gln Val Ala Glu Arg Met Ala Ile	
465	470 475 480
Gln Met Ala Leu Glu Asp Thr Arg Asp Lys Arg Val Asn Ile Val Thr	
485	490 495
Asp Ser Tyr Tyr Cys Trp Lys Asn Ile Thr Glu Gly Leu Gly Leu Glu	
500	505 510
Gly Pro Gln Ser Pro Trp Trp Pro Ile Ile Gln Asn Ile His Glu Lys	
515	520 525

-continued

Glu Ile Val Tyr Phe Ala Trp Val Pro Gly His Lys Gly Ile Cys Gly
530 535 540

Asn Gln Leu Ala Asp Glu Ala Ala Lys Ile Lys Glu Glu Ile
545 550 555

<210> SEQ ID NO 13
<211> LENGTH: 1674
<212> TYPE: DNA
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<221> NAME/KEY: source
<222> LOCATION: 1..1674
<223> OTHER INFORMATION: /mol_type="unassigned DNA"
/note="Synthetic gene"
/organism="Artificial Sequence"

<400> SEQUENCE: 13

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gttcagcgtc tgctgagcga aggtaaaatt agcgaagcaa gcgataataa cccgtataac      180
agcccgattt tcgtgatcaa aaaacgtagc ggtaaatggc gtctgctgca ggatctgcgt      240
gaactgaata aaaccgttca ggttggcacc gaaattagcc gtggtctgcc gcacccgggt      300
ggtctgatta aatgtaaaca catgaccgtt ctggatattg gcgacgcata tttaccatt      360
ccgtgggacg cggaatttcg tccgtatacc gcatttaca tccgagcat taatcatcaa      420
gaaccggata aacgttatgt gtggaattgc ctgccgcagg gttttgttct gagcccgat      480
atctatcaga aaacgctgca agaaattctg cagccatttc gtgaacgtta tccggaagtt      540
cagctgtatc agtatatgga tgacctgttt gtgggtagca atggtagcaa aaaacagcac      600
aaagaactga tcattgaact gcgtgcaatt ctgctggaaa aagggtttga aacaccggat      660
gataaactgc aagaagttcc gccttatagc tggtgggtt atcagctgtg tccggaaaat      720
tggaagttc agaaaatgca gctggatatg gttaaaaatc cgaccctgaa tgatgtgcag      780
aaactgggtg gcaaaattaa ttgggctagc caggggtgtc cgggtctgac cgttaaacat      840
attgcagcaa ccaccaaagg ttgtctggaa ctgaatcaga aagttatctg gaccgaagaa      900
gcacaaaaag aactggaaga aaacaacgag aaatcaaaa atgcacaggg cctgcagtat      960
tataaccggg aagaagaaat gctttgcgaa gtggaaatca ccaaaaacta tgaagccacc     1020
tatgtgatta aacagagcca gggatttctg tgggcaggca aaaaaggcat gaaagccgct     1080
aaaacccaaa gcaccgtgaa aaatctgatg ctgctgtgct aacatgtggc aaccgaaagc     1140
attaccctgt ttggtaaatg tccgaccttt aaagttccgt ttaccaaaga acaggtgatg     1200
tggaagtgc agaaaggttg gtattatagt tggctgccgg aaattgttta taccatcag     1260
gtgttcctat atgactggcg tatgaaactg gttgaagaac cgaccagcgg tattaccatt     1320
tataccgggt gtggtaaaca gaatggtgaa ggtattgcag cctatgttac cagcaatggt     1380
cgtaccaaac agaaacgtct gggtcgggtt acacatcagg tggcagaacg tatggcaatt     1440
cagatggcac tggaagatag ccgtgataaa cgggttaata ttgtgaccga tagctattat     1500
tgttggaaaa acattaccga aggtctgggt ctggaaggtc cgcagagccc gtggtggccg     1560
attattcaga atattcacga aaaagaaatc gtgtactttg catgggttcc aggtcataaa     1620
ggtatttgtg gtaatcagct ggcagatgaa gcagccaaaa tcaagaaga gatc      1674
```

<210> SEQ ID NO 14
<211> LENGTH: 558

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<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Expression from a synthetic gene

<400> SEQUENCE: 14

Met Ala Gln Leu Ser Lys Glu Ile Lys Phe Arg Lys Ile Glu Leu Lys
 1             5             10             15

Glu Gly Thr Met Gly Pro Lys Ile Pro Gln Trp Pro Leu Thr Lys Glu
          20             25             30

Lys Leu Glu Gly Ala Lys Glu Ile Val Gln Arg Leu Leu Ser Glu Gly
          35             40             45

Lys Ile Ser Glu Ala Ser Asp Asn Asn Pro Tyr Asn Ser Pro Ile Phe
 50             55             60

Val Ile Lys Lys Arg Ser Gly Lys Trp Arg Leu Leu Gln Asp Leu Arg
 65             70             75             80

Glu Leu Asn Lys Thr Val Gln Val Gly Thr Glu Ile Ser Arg Gly Leu
          85             90             95

Pro His Pro Gly Gly Leu Ile Lys Cys Lys His Met Thr Val Leu Asp
          100            105            110

Ile Gly Asp Ala Tyr Phe Thr Ile Pro Leu Asp Pro Glu Phe Arg Pro
          115            120            125

Tyr Thr Ala Phe Thr Ile Pro Ser Ile Asn His Gln Glu Pro Asp Lys
          130            135            140

Arg Tyr Val Trp Asn Cys Leu Pro Gln Gly Phe Val Leu Ser Pro Tyr
          145            150            155            160

Ile Tyr Gln Lys Thr Leu Gln Glu Ile Leu Gln Pro Phe Arg Glu Arg
          165            170            175

Tyr Pro Glu Val Gln Leu Tyr Gln Tyr Met Asp Asp Leu Phe Val Gly
          180            185            190

Ser Asn Gly Ser Lys Lys Gln His Lys Glu Leu Ile Ile Glu Leu Arg
          195            200            205

Ala Ile Leu Leu Glu Lys Gly Phe Glu Thr Pro Asp Asp Lys Leu Gln
          210            215            220

Glu Val Pro Pro Tyr Ser Trp Leu Gly Tyr Gln Leu Cys Pro Glu Asn
          225            230            235            240

Trp Lys Val Gln Lys Met Gln Leu Asp Met Val Lys Asn Pro Thr Leu
          245            250            255

Asn Asp Val Gln Lys Leu Val Gly Lys Ile Asn Trp Ala Ser Gln Gly
          260            265            270

Val Pro Gly Leu Thr Val Lys His Ile Ala Ala Thr Thr Lys Gly Cys
          275            280            285

Leu Glu Leu Asn Gln Lys Val Ile Trp Thr Glu Glu Ala Gln Lys Glu
          290            295            300

Leu Glu Glu Asn Asn Glu Lys Ile Lys Asn Ala Gln Gly Leu Gln Tyr
          305            310            315            320

Tyr Asn Pro Glu Glu Glu Met Leu Cys Glu Val Glu Ile Thr Lys Asn
          325            330            335

Tyr Glu Ala Thr Tyr Val Ile Lys Gln Ser Gln Gly Ile Leu Trp Ala
          340            345            350

Gly Lys Lys Gly Met Lys Ala Ala Lys Thr Lys Ser Thr Val Lys Asn
          355            360            365

Leu Met Leu Leu Leu Gln His Val Ala Thr Glu Ser Ile Thr Arg Val
          370            375            380

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Gly	Lys	Cys	Pro	Thr	Phe	Lys	Val	Pro	Phe	Thr	Lys	Glu	Gln	Val	Met	385	390	395	400
Trp	Glu	Met	Gln	Lys	Gly	Trp	Tyr	Tyr	Ser	Trp	Leu	Pro	Glu	Ile	Val	405	410	415	
Tyr	Thr	His	Gln	Val	Val	His	Asp	Asp	Trp	Arg	Met	Lys	Leu	Val	Glu	420	425	430	
Glu	Pro	Thr	Ser	Gly	Ile	Thr	Ile	Tyr	Thr	Gly	Gly	Gly	Lys	Gln	Asn	435	440	445	
Gly	Glu	Gly	Ile	Ala	Ala	Tyr	Val	Thr	Ser	Asn	Gly	Arg	Thr	Lys	Gln	450	455	460	
Lys	Arg	Leu	Gly	Pro	Val	Thr	His	Gln	Val	Ala	Glu	Arg	Met	Ala	Ile	465	470	475	480
Gln	Met	Ala	Leu	Glu	Asp	Thr	Arg	Asp	Lys	Arg	Val	Asn	Ile	Val	Thr	485	490	495	
Asp	Ser	Tyr	Tyr	Cys	Trp	Lys	Asn	Ile	Thr	Glu	Gly	Leu	Gly	Leu	Glu	500	505	510	
Gly	Pro	Gln	Ser	Pro	Trp	Trp	Pro	Ile	Ile	Gln	Asn	Ile	His	Glu	Lys	515	520	525	
Glu	Ile	Val	Tyr	Phe	Ala	Trp	Val	Pro	Gly	His	Lys	Gly	Ile	Cys	Gly	530	535	540	
Asn	Gln	Leu	Ala	Asp	Glu	Ala	Ala	Lys	Ile	Lys	Glu	Glu	Ile			545	550	555	

<210> SEQ ID NO 15

<211> LENGTH: 558

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Expression from a synthetic gene

<400> SEQUENCE: 15

Met	Ala	Gln	Leu	Ser	Lys	Glu	Ile	Lys	Phe	Arg	Lys	Ile	Glu	Leu	Lys	1	5	10	15
Glu	Gly	Thr	Met	Gly	Pro	Lys	Ile	Pro	Gln	Trp	Pro	Leu	Thr	Lys	Glu	20	25	30	
Lys	Leu	Glu	Gly	Ala	Lys	Glu	Ile	Val	Gln	Arg	Leu	Leu	Ser	Glu	Gly	35	40	45	
Lys	Ile	Ser	Glu	Ala	Ser	Asp	Asn	Asn	Pro	Tyr	Asn	Ser	Pro	Ile	Phe	50	55	60	
Val	Ile	Lys	Lys	Arg	Ser	Gly	Lys	Trp	Arg	Leu	Leu	Gln	Asp	Leu	Arg	65	70	75	80
Glu	Leu	Asn	Lys	Thr	Val	Gln	Val	Gly	Thr	Glu	Ile	Ser	Arg	Gly	Leu	85	90	95	
Pro	His	Pro	Gly	Gly	Leu	Ile	Lys	Cys	Lys	His	Met	Thr	Val	Leu	Asp	100	105	110	
Ile	Gly	Asp	Ala	Tyr	Phe	Thr	Ile	Pro	Leu	Asp	Pro	Glu	Phe	Arg	Pro	115	120	125	
Tyr	Thr	Ala	Phe	Thr	Ile	Pro	Ser	Ile	Asn	His	Gln	Glu	Pro	Asp	Lys	130	135	140	
Arg	Tyr	Val	Trp	Asn	Cys	Leu	Pro	Gln	Gly	Phe	Val	Leu	Ser	Pro	Tyr	145	150	155	160
Ile	Tyr	Gln	Lys	Thr	Leu	Gln	Glu	Ile	Leu	Gln	Pro	Phe	Arg	Glu	Arg	165	170	175	
Tyr	Pro	Glu	Val	Gln	Leu	Tyr	Gln	Tyr	Met	Asp	Asp	Leu	Phe	Val	Gly	180	185	190	

Ser	Asn	Gly	Ser	Lys	Lys	Gln	His	Lys	Glu	Leu	Ile	Ile	Glu	Leu	Arg
195						200			205						
Ala	Ile	Leu	Leu	Glu	Lys	Gly	Phe	Glu	Thr	Pro	Asp	Asp	Lys	Leu	Gln
210						215			220						
Glu	Val	Pro	Pro	Tyr	Ser	Trp	Leu	Gly	Tyr	Gln	Leu	Cys	Pro	Glu	Asn
225						230			235			240			
Trp	Lys	Val	Gln	Lys	Met	Gln	Leu	Asp	Met	Val	Lys	Asn	Pro	Thr	Leu
			245						250			255			
Asn	Asp	Val	Gln	Lys	Leu	Val	Gly	Lys	Ile	Asn	Trp	Ala	Ser	Gln	Gly
			260						265			270			
Val	Pro	Gly	Leu	Thr	Val	Lys	His	Ile	Ala	Ala	Thr	Thr	Lys	Gly	Cys
275						280						285			
Leu	Glu	Leu	Asn	Gln	Lys	Val	Ile	Trp	Thr	Glu	Glu	Ala	Gln	Lys	Glu
290						295						300			
Leu	Glu	Glu	Asn	Asn	Glu	Lys	Ile	Lys	Asn	Ala	Gln	Gly	Leu	Gln	Tyr
305						310			315			320			
Tyr	Asn	Pro	Glu	Glu	Glu	Met	Leu	Cys	Glu	Val	Glu	Ile	Thr	Lys	Asn
			325						330			335			
Tyr	Glu	Ala	Thr	Tyr	Val	Ile	Lys	Gln	Ser	Gln	Gly	Ile	Leu	Trp	Ala
			340						345			350			
Gly	Lys	Lys	Gly	Met	Lys	Ala	Ala	Lys	Thr	Asn	Ser	Thr	Val	Lys	Asn
355						360						365			
Leu	Met	Leu	Leu	Leu	Gln	His	Val	Ala	Thr	Glu	Ser	Ile	Thr	Arg	Val
370						375						380			
Gly	Lys	Cys	Pro	Thr	Phe	Lys	Val	Pro	Phe	Thr	Lys	Glu	Gln	Val	Met
385						390			395			400			
Trp	Glu	Met	Gln	Lys	Gly	Trp	Tyr	Tyr	Ser	Trp	Leu	Pro	Glu	Ile	Val
			405						410			415			
Tyr	Thr	His	Gln	Val	Val	His	Asp	Asp	Trp	Arg	Met	Lys	Leu	Val	Glu
			420						425			430			
Glu	Pro	Thr	Ser	Gly	Ile	Thr	Ile	Tyr	Thr	Gly	Gly	Gly	Lys	Gln	Asn
435						440						445			
Gly	Glu	Gly	Ile	Ala	Ala	Tyr	Val	Thr	Ser	Asn	Gly	Arg	Thr	Lys	Gln
450						455						460			
Lys	Arg	Leu	Gly	Pro	Val	Thr	His	Gln	Val	Ala	Glu	Arg	Met	Ala	Ile
465						470			475			480			
Gln	Met	Ala	Leu	Glu	Asp	Thr	Arg	Asp	Lys	Gln	Val	Asn	Ile	Val	Thr
			485						490			495			
Asp	Ser	Tyr	Tyr	Cys	Trp	Lys	Asn	Ile	Thr	Glu	Gly	Leu	Gly	Leu	Glu
			500						505			510			
Gly	Pro	Gln	Ser	Pro	Trp	Trp	Pro	Ile	Ile	Gln	Asn	Ile	Arg	Glu	Lys
515						520						525			
Glu	Ile	Val	Tyr	Phe	Ala	Trp	Val	Pro	Gly	His	Lys	Gly	Ile	Cys	Gly
530						535						540			
Asn	Gln	Leu	Ala	Asp	Glu	Ala	Ala	Lys	Ile	Lys	Glu	Glu	Ile		
545			550						555						

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<210> SEQ ID NO 16
<211> LENGTH: 558
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Expression from a synthetic gene

<400> SEQUENCE: 16
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Met	Ala	Gln	Leu	Ser	Lys	Glu	Ile	Lys	Phe	Arg	Lys	Ile	Glu	Leu	Lys	1	5	10	15
Glu	Gly	Thr	Met	Gly	Pro	Lys	Ile	Pro	Gln	Trp	Pro	Leu	Thr	Lys	Glu	20	25	30	
Lys	Leu	Glu	Gly	Ala	Lys	Glu	Ile	Val	Gln	Arg	Leu	Leu	Ser	Glu	Gly	35	40	45	
Lys	Ile	Ser	Glu	Ala	Ser	Asp	Asn	Asn	Pro	Tyr	Asn	Ser	Pro	Ile	Phe	50	55	60	
Val	Ile	Lys	Lys	Arg	Ser	Gly	Lys	Trp	Arg	Leu	Leu	Gln	Asp	Leu	Arg	65	70	75	80
Glu	Leu	Asn	Lys	Thr	Val	Gln	Val	Gly	Thr	Glu	Ile	Ser	Arg	Gly	Leu	85	90	95	
Pro	His	Pro	Gly	Gly	Leu	Ile	Lys	Cys	Lys	His	Met	Thr	Val	Leu	Asp	100	105	110	
Ile	Gly	Asp	Ala	Tyr	Phe	Thr	Ile	Pro	Leu	Asp	Pro	Glu	Phe	Arg	Pro	115	120	125	
Tyr	Thr	Ala	Phe	Thr	Ile	Pro	Ser	Ile	Asn	His	Gln	Glu	Pro	Asp	Lys	130	135	140	
Arg	Tyr	Val	Trp	Asn	Cys	Leu	Pro	Gln	Gly	Phe	Val	Leu	Ser	Pro	Tyr	145	150	155	160
Ile	Tyr	Gln	Lys	Thr	Leu	Gln	Glu	Ile	Leu	Gln	Pro	Phe	Arg	Glu	Arg	165	170	175	
Tyr	Pro	Glu	Val	Gln	Leu	Tyr	Gln	Tyr	Met	Asp	Asp	Leu	Phe	Val	Gly	180	185	190	
Ser	Asn	Gly	Ser	Lys	Lys	Gln	His	Lys	Glu	Leu	Ile	Ile	Glu	Leu	Arg	195	200	205	
Ala	Ile	Leu	Leu	Glu	Lys	Gly	Phe	Glu	Thr	Pro	Asp	Asp	Lys	Leu	Gln	210	215	220	
Glu	Val	Pro	Pro	Tyr	Ser	Trp	Leu	Gly	Tyr	Gln	Leu	Cys	Pro	Glu	Asn	225	230	235	240
Trp	Lys	Val	Gln	Lys	Met	Gln	Leu	Asp	Met	Val	Lys	Asn	Pro	Thr	Leu	245	250	255	
Asn	Asp	Val	Gln	Lys	Leu	Val	Gly	Lys	Ile	Asn	Trp	Ala	Ser	Gln	Gly	260	265	270	
Val	Pro	Gly	Leu	Thr	Val	Lys	His	Ile	Ala	Ala	Thr	Thr	Lys	Gly	Cys	275	280	285	
Leu	Glu	Leu	Asn	Gln	Lys	Val	Ile	Trp	Thr	Glu	Glu	Ala	Gln	Lys	Glu	290	295	300	
Leu	Glu	Glu	Asn	Asn	Glu	Lys	Ile	Lys	Asn	Ala	Gln	Gly	Leu	Gln	Tyr	305	310	315	320
Tyr	Asn	Pro	Glu	Glu	Glu	Met	Leu	Cys	Glu	Val	Glu	Ile	Thr	Lys	Asn	325	330	335	
Tyr	Glu	Ala	Thr	Tyr	Val	Ile	Lys	Gln	Ser	Gln	Gly	Ile	Leu	Trp	Ala	340	345	350	
Gly	Lys	Lys	Asp	Met	Lys	Ala	Ala	Lys	Thr	Lys	Ser	Thr	Val	Lys	Asn	355	360	365	
Leu	Met	Leu	Leu	Leu	Gln	His	Val	Ala	Thr	Glu	Ser	Ile	Thr	Arg	Val	370	375	380	
Gly	Lys	Cys	Pro	Thr	Phe	Lys	Val	Pro	Phe	Thr	Lys	Glu	Gln	Val	Met	385	390	395	400
Trp	Glu	Met	Gln	Lys	Gly	Trp	Tyr	Tyr	Ser	Trp	Leu	Pro	Glu	Ile	Val	405	410	415	
Tyr	Thr	His	Gln	Val	Val	His	Asp	Asp	Trp	Arg	Met	Lys	Leu	Val	Glu				

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420	425	430
Glu Pro Thr Ser Gly Ile Thr Ile Tyr Thr Gly Gly Gly Lys Gln Asn		
435	440	445
Gly Glu Gly Ile Ala Ala Tyr Val Thr Ser Asn Gly Arg Thr Lys Gln		
450	455	460
Lys Arg Leu Gly Pro Val Thr His Gln Val Ala Glu Arg Met Ala Ile		
465	470	475
Gln Met Ala Leu Glu Asp Thr Arg Asp Lys Gln Val Asn Ile Val Thr		
485	490	495
Asp Ser Tyr Tyr Cys Trp Lys Asn Ile Thr Glu Gly Leu Gly Leu Glu		
500	505	510
Gly Pro Gln Ser Pro Trp Trp Pro Ile Ile Gln Asn Ile Arg Glu Lys		
515	520	525
Glu Ile Val Tyr Phe Ala Trp Val Pro Gly His Lys Gly Ile Cys Gly		
530	535	540
Asn Gln Leu Ala Asp Glu Ala Ala Lys Ile Lys Glu Glu Ile		
545	550	555

<210> SEQ ID NO 17
 <211> LENGTH: 558
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Expression from a synthetic gene

<400> SEQUENCE: 17

Met Ala Gln Leu Ser Lys Glu Ile Lys Phe Arg Lys Ile Glu Leu Lys		
1	5	10
Glu Gly Thr Met Gly Pro Lys Ile Pro Gln Trp Pro Leu Thr Lys Glu		
20	25	30
Lys Leu Glu Gly Ala Lys Glu Ile Val Gln Arg Leu Leu Ser Glu Gly		
35	40	45
Lys Ile Ser Glu Ala Ser Asp Asn Asn Pro Tyr Asn Ser Pro Ile Phe		
50	55	60
Val Ile Lys Lys Arg Ser Gly Lys Trp Arg Leu Leu Gln Asp Leu Arg		
65	70	75
Glu Leu Asn Lys Thr Val Gln Val Gly Thr Glu Ile Ser Arg Gly Leu		
85	90	95
Pro His Pro Gly Gly Leu Ile Lys Cys Lys His Met Thr Val Leu Asp		
100	105	110
Ile Gly Asp Ala Tyr Phe Thr Ile Pro Leu Asp Pro Glu Phe Arg Pro		
115	120	125
Tyr Thr Ala Phe Thr Ile Pro Ser Ile Asn His Gln Glu Pro Asp Lys		
130	135	140
Arg Tyr Val Trp Asn Cys Leu Pro Gln Gly Phe Val Leu Ser Pro Tyr		
145	150	155
Ile Tyr Gln Lys Thr Leu Gln Glu Ile Leu Gln Pro Phe Arg Glu Arg		
165	170	175
Tyr Pro Glu Val Gln Leu Tyr Gln Tyr Met Asp Asp Leu Phe Val Gly		
180	185	190
Ser Asn Gly Ser Lys Lys Gln His Lys Glu Leu Ile Ile Glu Leu Arg		
195	200	205
Ala Ile Leu Leu Glu Lys Gly Phe Glu Thr Pro Asp Asp Lys Leu Gln		
210	215	220
Glu Val Pro Pro Tyr Ser Trp Leu Gly Tyr Gln Leu Cys Pro Glu Asn		

225	230						235						240			
Trp	Lys	Val	Gln	Lys	Met	Gln	Leu	Asp	Met	Val	Lys	Asn	Pro	Thr	Leu	
				245					250					255		
Asn	Asp	Val	Gln	Lys	Leu	Val	Gly	Lys	Ile	Asn	Trp	Ala	Ser	Gln	Gly	
				260					265					270		
Val	Pro	Gly	Leu	Thr	Val	Lys	His	Ile	Ala	Ala	Thr	Thr	Lys	Gly	Cys	
				275					280					285		
Leu	Glu	Leu	Asn	Gln	Lys	Val	Ile	Trp	Thr	Glu	Glu	Ala	Gln	Lys	Glu	
				290					295					300		
Leu	Glu	Glu	Asn	Asn	Glu	Lys	Ile	Lys	Asn	Ala	Gln	Gly	Leu	Gln	Tyr	
				305					310					315		
Tyr	Asn	Pro	Glu	Glu	Glu	Met	Leu	Cys	Glu	Val	Glu	Ile	Thr	Lys	Asn	
				325					330					335		
Tyr	Glu	Ala	Thr	Tyr	Val	Ile	Lys	Gln	Ser	Gln	Gly	Ile	Leu	Trp	Ala	
				340					345					350		
Gly	Lys	Lys	Asn	Met	Lys	Ala	Ala	Lys	Asn	Ile	Ser	Thr	Val	Lys	Asn	
				355					360					365		
Leu	Met	Leu	Leu	Leu	Gln	His	Val	Ala	Thr	Glu	Ser	Ile	Thr	Arg	Val	
				370					375					380		
Gly	Lys	Cys	Pro	Thr	Phe	Lys	Val	Pro	Phe	Thr	Lys	Glu	Gln	Val	Met	
				385					390					395		
Trp	Glu	Met	Gln	Lys	Gly	Trp	Tyr	Tyr	Ser	Trp	Leu	Pro	Glu	Ile	Val	
				405					410					415		
Tyr	Thr	His	Gln	Val	Val	His	Asp	Asp	Trp	Arg	Met	Lys	Leu	Val	Glu	
				420					425					430		
Glu	Pro	Thr	Ser	Gly	Ile	Thr	Ile	Tyr	Thr	Gly	Gly	Gly	Lys	Gln	Asn	
				435					440					445		
Gly	Glu	Gly	Ile	Ala	Ala	Tyr	Val	Thr	Ser	Asn	Gly	Arg	Thr	Lys	Gln	
				450					455					460		
Lys	Arg	Leu	Gly	Pro	Val	Thr	His	Gln	Val	Ala	Glu	Arg	Met	Ala	Ile	
				465					470					475		
Gln	Met	Ala	Leu	Glu	Asp	Thr	Arg	Asp	Lys	Gln	Val	Asn	Ile	Val	Thr	
				485					490					495		
Asp	Ser	Tyr	Tyr	Cys	Trp	Lys	Asn	Ile	Thr	Glu	Gly	Leu	Gly	Leu	Glu	
				500					505					510		
Gly	Pro	Gln	Ser	Pro	Trp	Trp	Pro	Ile	Ile	Gln	Asn	Ile	Arg	Glu	Lys	
				515					520					525		
Glu	Ile	Val	Tyr	Phe	Ala	Trp	Val	Pro	Gly	His	Lys	Gly	Ile	Cys	Gly	
				530					535					540		
Asn	Gln	Leu	Ala	Asp	Glu	Ala	Ala	Lys	Ile	Lys	Glu	Glu	Ile			
				545					550					555		

<400> SEQUENCE: 18

Lys Leu Glu Gly Ala Lys Glu Ile Val Gln Arg Leu Leu Ser Glu Gly

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35					40					45					
Lys	Ile	Ser	Glu	Ala	Ser	Asp	Asn	Asn	Pro	Tyr	Asn	Ser	Pro	Ile	Phe
50						55					60				
Val	Ile	Lys	Lys	Arg	Ser	Gly	Lys	Trp	Arg	Leu	Leu	Gln	Asp	Leu	Arg
65					70					75					80
Glu	Leu	Asn	Lys	Thr	Val	Gln	Val	Gly	Thr	Glu	Ile	Ser	Arg	Gly	Leu
				85					90					95	
Pro	His	Pro	Gly	Gly	Leu	Ile	Lys	Cys	Lys	His	Met	Thr	Val	Leu	Asp
			100					105					110		
Ile	Gly	Asp	Ala	Tyr	Phe	Thr	Ile	Pro	Leu	Asp	Pro	Glu	Phe	Arg	Pro
		115					120					125			
Tyr	Thr	Ala	Phe	Thr	Ile	Pro	Ser	Ile	Asn	His	Gln	Glu	Pro	Asp	Lys
	130					135					140				
Arg	Tyr	Val	Trp	Asn	Cys	Leu	Pro	Gln	Gly	Phe	Val	Leu	Ser	Pro	Tyr
145					150					155					160
Ile	Tyr	Gln	Lys	Thr	Leu	Gln	Glu	Ile	Leu	Gln	Pro	Phe	Arg	Glu	Arg
			165					170						175	
Tyr	Pro	Glu	Val	Gln	Leu	Tyr	Gln	Tyr	Met	Asp	Asp	Leu	Phe	Val	Gly
		180						185					190		
Ser	Asn	Gly	Ser	Lys	Lys	Gln	His	Lys	Glu	Leu	Ile	Ile	Glu	Leu	Arg
	195						200					205			
Ala	Ile	Leu	Leu	Glu	Lys	Gly	Phe	Glu	Thr	Pro	Asp	Asp	Lys	Leu	Gln
	210					215					220				
Glu	Val	Pro	Pro	Tyr	Ser	Trp	Leu	Gly	Tyr	Gln	Leu	Cys	Pro	Glu	Asn
225					230					235					240
Trp	Lys	Val	Gln	Lys	Met	Gln	Leu	Asp	Met	Val	Lys	Asn	Pro	Thr	Leu
			245						250					255	
Asn	Asp	Val	Gln	Lys	Leu	Val	Gly	Lys	Ile	Asn	Trp	Ala	Ser	Gln	Gly
		260						265					270		
Val	Pro	Gly	Leu	Thr	Val	Lys	His	Ile	Ala	Ala	Thr	Thr	Lys	Gly	Cys
		275					280						285		
Leu	Glu	Leu	Asn	Gln	Lys	Val	Ile	Trp	Thr	Glu	Glu	Ala	Gln	Lys	Glu
	290					295					300				
Leu	Glu	Glu	Asn	Asn	Glu	Lys	Ile	Lys	Asn	Ala	Gln	Gly	Leu	Gln	Tyr
305					310					315					320
Tyr	Asn	Pro	Glu	Glu	Glu	Met	Leu	Cys	Glu	Val	Glu	Ile	Thr	Lys	Asn
			325						330					335	
Tyr	Glu	Ala	Thr	Tyr	Val	Ile	Lys	Gln	Ser	Gln	Gly	Ile	Leu	Trp	Ala
		340						345					350		
Gly	Lys	Lys	Ile	Met	Lys	Ala	Asn	Lys	Gly	Trp	Ser	Thr	Val	Lys	Asn
		355					360						365		
Leu	Met	Leu	Leu	Leu	Gln	His	Val	Ala	Thr	Glu	Ser	Ile	Thr	Arg	Val
	370					375					380				
Gly	Lys	Cys	Pro	Thr	Phe	Lys	Val	Pro	Phe	Thr	Lys	Glu	Gln	Val	Met
385					390					395					400
Trp	Glu	Met	Gln	Lys	Gly	Trp	Tyr	Tyr	Ser	Trp	Leu	Pro	Glu	Ile	Val
			405						410					415	
Tyr	Thr	His	Gln	Val	Val	His	Asp	Asp	Trp	Arg	Met	Lys	Leu	Val	Glu
			420					425					430		
Glu	Pro	Thr	Ser	Gly	Ile	Thr	Ile	Tyr	Thr	Gly	Gly	Gly	Lys	Gln	Asn
		435					440					445			
Gly	Glu	Gly	Ile	Ala	Ala	Tyr	Val	Thr	Ser	Asn	Gly	Arg	Thr	Lys	Gln
	450					455					460				

-continued

Lys Arg Leu Gly Pro Val Thr His Gln Val Ala Glu Arg Met Ala Ile
 465 470 475 480
 Gln Met Ala Leu Glu Asp Thr Arg Asp Lys Arg Val Asn Ile Val Thr
 485 490 495
 Asp Ser Tyr Tyr Cys Trp Lys Asn Ile Thr Glu Gly Leu Gly Leu Glu
 500 505 510
 Gly Pro Gln Ser Pro Trp Trp Pro Ile Ile Gln Asn Ile Arg Glu Lys
 515 520 525
 Glu Ile Val Tyr Phe Ala Trp Val Pro Gly His Lys Gly Ile Cys Gly
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 Asn Gln Leu Ala Asp Glu Ala Ala Arg Ile Lys Glu Glu Ile
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<210> SEQ ID NO 19
 <211> LENGTH: 558
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Expression from a synthetic gene

<400> SEQUENCE: 19

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 Glu Gly Thr Met Gly Pro Lys Ile Pro Gln Trp Pro Leu Thr Lys Glu
 20 25 30
 Lys Leu Glu Gly Ala Lys Glu Ile Val Gln Arg Leu Leu Ser Glu Gly
 35 40 45
 Lys Ile Ser Glu Ala Ser Asp Asn Asn Pro Tyr Asn Ser Pro Ile Phe
 50 55 60
 Val Ile Lys Lys Arg Ser Gly Lys Trp Arg Leu Leu Gln Asp Leu Arg
 65 70 75 80
 Glu Leu Asn Lys Thr Val Gln Val Gly Thr Glu Ile Ser Arg Gly Leu
 85 90 95
 Pro His Pro Gly Gly Leu Ile Lys Cys Lys His Met Thr Val Leu Asp
 100 105 110
 Ile Gly Asp Ala Tyr Phe Thr Ile Pro Leu Asp Pro Glu Phe Arg Pro
 115 120 125
 Tyr Thr Ala Phe Thr Ile Pro Ser Ile Asn His Gln Glu Pro Asp Lys
 130 135 140
 Arg Tyr Val Trp Asn Cys Leu Pro Gln Gly Phe Val Leu Ser Pro Tyr
 145 150 155 160
 Ile Tyr Gln Lys Thr Leu Gln Glu Ile Leu Gln Pro Phe Arg Glu Arg
 165 170 175
 Tyr Pro Glu Val Gln Leu Tyr Gln Tyr Met Asp Asp Leu Phe Val Gly
 180 185 190
 Ser Asn Gly Ser Lys Lys Gln His Lys Glu Leu Ile Ile Glu Leu Arg
 195 200 205
 Ala Ile Leu Leu Glu Lys Gly Phe Glu Thr Pro Asp Asp Lys Leu Gln
 210 215 220
 Glu Val Pro Pro Tyr Ser Trp Leu Gly Tyr Gln Leu Cys Pro Glu Asn
 225 230 235 240
 Trp Lys Val Gln Lys Met Gln Leu Asp Met Val Lys Asn Pro Thr Leu
 245 250 255
 Asn Asp Val Gln Lys Leu Val Gly Lys Ile Asn Trp Ala Ser Gln Gly
 260 265 270

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Val	Pro	Gly	Leu	Thr	Val	Lys	His	Ile	Ala	Ala	Thr	Thr	Lys	Gly	Cys
		275					280					285			
Leu	Glu	Leu	Asn	Gln	Lys	Val	Ile	Trp	Thr	Glu	Glu	Ala	Gln	Lys	Glu
		290				295					300				
Leu	Glu	Glu	Asn	Asn	Glu	Lys	Ile	Lys	Asn	Ala	Gln	Gly	Leu	Gln	Tyr
		305			310					315					320
Tyr	Asn	Pro	Glu	Glu	Glu	Met	Leu	Cys	Glu	Val	Glu	Ile	Thr	Lys	Asn
			325						330					335	
Tyr	Glu	Ala	Thr	Tyr	Val	Ile	Lys	Gln	Ser	Gln	Gly	Ile	Leu	Trp	Ala
			340					345					350		
Gly	Lys	Lys	Ile	Met	Lys	Ala	Asn	Lys	Gly	Trp	Ser	Thr	Val	Lys	Asn
		355					360					365			
Leu	Met	Leu	Leu	Leu	Gln	His	Val	Ala	Thr	Glu	Ser	Ile	Thr	Arg	Val
		370				375					380				
Gly	Lys	Cys	Pro	Thr	Phe	Lys	Val	Pro	Phe	Thr	Lys	Glu	Gln	Val	Met
		385			390					395					400
Trp	Glu	Met	Gln	Lys	Gly	Trp	Tyr	Tyr	Ser	Trp	Leu	Pro	Glu	Ile	Val
			405						410					415	
Tyr	Thr	His	Gln	Val	Val	His	Asp	Asp	Trp	Arg	Met	Lys	Leu	Val	Glu
			420				425						430		
Glu	Pro	Thr	Ser	Gly	Ile	Thr	Ile	Tyr	Thr	Gly	Gly	Gly	Lys	Gln	Asn
		435				440					445				
Gly	Glu	Gly	Ile	Ala	Ala	Tyr	Val	Thr	Ser	Asn	Gly	Arg	Thr	Lys	Gln
		450				455					460				
Lys	Arg	Leu	Gly	Pro	Phe	Thr	His	Gln	Val	Ala	Glu	Arg	Met	Ala	Ile
		465			470					475					480
Gln	Met	Ala	Leu	Glu	Asp	Thr	Arg	Asp	Lys	Arg	Val	Asn	Ile	Val	Thr
			485					490						495	
Asp	Ser	Tyr	Tyr	Cys	Trp	Lys	Asn	Ile	Thr	Glu	Gly	Leu	Gly	Leu	Glu
			500				505						510		
Gly	Pro	Gln	Ser	Pro	Trp	Trp	Pro	Ile	Ile	Gln	Asn	Ile	His	Glu	Lys
		515					520					525			
Glu	Ile	Val	Tyr	Phe	Ala	Trp	Val	Pro	Gly	His	Lys	Gly	Ile	Cys	Gly
		530				535					540				
Asn	Gln	Leu	Ala	Asp	Glu	Ala	Ala	Arg	Ile	Lys	Glu	Glu	Ile		
		545			550				555						

The invention claimed is:

1. A reverse transcriptase comprising two subunits, wherein the two subunits are each encoded by a variant of the polynucleotide sequence of SEQ ID NO:1, wherein the amino acid sequence encoded by the variant is at least 90% identical to the amino acid sequence of SEQ ID NO: 2, wherein the variant comprises mutations in the polynucleotide sequence of SEQ ID NO:1 causing amino acid exchanges relative to the amino acid sequence of SEQ ID NO:2 in the minor groove binding track (MGBT) of the reverse transcriptase's thumb domain, and wherein the mutations in the polynucleotide sequence of SEQ ID NO:1 cause amino acid exchanges within the encoded amino acid sequence at all the amino acid positions that correspond to positions 263, 265, 267, 269, and 271 of SEQ ID NO:2, wherein the amino acid exchanges are M263V, N265K, T267N, M269A, and S271Q.

2. The reverse transcriptase according to claim 1, wherein the variant comprises one or more further mutations in the

polynucleotide sequence of SEQ ID NO:1 causing one or more further amino acid exchanges relative to SEQ ID NO: 2 in the reverse transcriptase's connection domain.

3. The reverse transcriptase according to claim 2, wherein the one or more further mutations in the polynucleotide sequence of SEQ ID NO: 1 cause one or more further amino acid exchanges within the encoded amino acid sequence at the amino acid positions that correspond to positions 356, 360, 362, and 363 of SEQ ID NO:2, wherein the amino acid exchanges are I356G/I356D/I356N and/or N360A and/or G362T/G362N and/or W363K/W363N/W363I.

4. The reverse transcriptase according to claim 2, wherein the one or more further mutations in the polynucleotide sequence of SEQ ID NO: 1 cause one or more further amino acid exchanges within the encoded amino acid sequence at the amino acid positions that correspond to positions 356, 360, 362, and 363 of SEQ ID NO:2, wherein the amino acid exchanges are I356G and/or N360A and/or G362T and/or W363K.

71

5. The reverse transcriptase according to claim 2, wherein the one or more further mutations in the polynucleotide sequence of SEQ ID NO: 1 cause one or more further amino acid exchanges within the encoded amino acid sequence at the amino acid positions that correspond to positions 356, 360, 362, and 363 of SEQ ID NO:2, wherein the amino acid exchanges are 1356G, N360A, G362T and W363K.

6. The reverse transcriptase according to claim 1, wherein the variant comprises one or more further mutations in the polynucleotide sequence of SEQ ID NO:1 causing one or more further amino acid exchanges relative to SEQ ID NO: 2 in the reverse transcriptase's RNase H domain.

7. The reverse transcriptase according to claim 6, wherein the further mutations in the polynucleotide sequence of SEQ ID NO: 1 cause one or more further amino acid exchanges within the encoded amino acid sequence at the amino acid positions that correspond to positions 443 and 476 of SEQ ID NO:2, wherein the amino acid exchanges are D443G and/or E476Q.

8. The reverse transcriptase according to claim 6, wherein the further mutations in the polynucleotide sequence of SEQ ID NO: 1 cause a further amino acid exchange within the encoded amino acid sequence at the amino acid position that corresponds to position 443 of SEQ ID NO:2, wherein the amino acid exchange is D443G.

9. The reverse transcriptase according to claim 1, wherein the variant comprises a further mutation in the polynucleotide sequence of SEQ ID NO:1 causing a further amino acid exchange relative to SEQ ID NO:2, wherein the further mutation in the polynucleotide sequence of SEQ ID NO:1 causes a further amino acid exchange within the encoded amino acid sequence at the amino acid position that corresponds to position 491 of SEQ ID NO:2, wherein the amino acid exchange is Q491R.

10. The reverse transcriptase according to claim 9, wherein the variant comprises one or more further mutations in the polynucleotide sequence of SEQ ID NO: 1 causing one or more further amino acid exchanges relative to SEQ ID NO:2, wherein the further mutations in the polynucleotide sequence of SEQ ID NO:1 cause one or more further amino acid exchanges within the encoded amino acid sequence at the amino acid positions that correspond to positions 470, 526 and 553 of SEQ ID NO:2, wherein the amino acid exchanges are V470F and/or R526H and/or K553R.

11. The reverse transcriptase according to claim 9, wherein the variant comprises one or more further mutations in the polynucleotide sequence of SEQ ID NO: 1 causing one or more further amino acid exchanges relative to SEQ ID NO:2, wherein the further mutations in the polynucleotide sequence of SEQ ID NO:1 cause one or more further

72

amino acid exchanges within the encoded amino acid sequence at the amino acid positions that correspond to positions 526 and 553 of SEQ ID NO:2, wherein the amino acid exchanges are R526H and/or K553R.

12. The reverse transcriptase according to claim 9, wherein the variant comprises one or more further mutations in the polynucleotide sequence of SEQ ID NO: 1 causing one or more further amino acid exchanges relative to SEQ ID NO:2, wherein the further mutations in the polynucleotide sequence of SEQ ID NO:1 cause a further amino acid exchange within the encoded amino acid sequence at the amino acid position that corresponds to position 526 of SEQ ID NO:2, wherein the amino acid exchange is R526H.

13. The reverse transcriptase according to claim 1, wherein the mutations in the polynucleotide sequence of SEQ ID NO: 1 cause amino acid exchanges within the encoded amino acid sequence at all the amino acid positions that correspond to positions 263, 265, 267, 269, 271, 356, 360, 362, 363 and 443 of SEQ ID NO:2, wherein the amino acid exchanges are M263V, N265K, T267N, M269A, S271Q, I356G, N360A, G362T, W363K and D443G.

14. The reverse transcriptase according to claim 1, wherein a first subunit of the two subunits comprises the complete amino acid sequence encoded by the variant and wherein a second subunit of the two subunits comprises a proteolytic fragment of the complete amino acid sequence encoded by the variant.

15. A kit comprising:

- a) the reverse transcriptase of claim 1; and
- b) a buffer.

16. A polynucleotide encoding an reverse transcriptase according to claim 1.

17. A vector comprising the polynucleotide according to claim 16.

18. A transformed host cell comprising the vector according to claim 17.

19. A reverse transcriptase obtained by expression of the polynucleotide according to claim 16 in a host cell.

20. The reverse transcriptase according to claim 19, wherein the host cell is *E. coli*.

21. A reverse transcriptase obtained by expression of the vector according to claim 17 in a host cell.

22. The reverse transcriptase according to claim 21, wherein the host cell is *E. coli*.

23. A method for amplifying template nucleic acids comprising contacting the template nucleic acids with an reverse transcriptase according to claim 1.

24. The method according to claim 23, wherein the method is reverse transcription (RT) PCR.

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